

29th Oct

2022

TOPIC

1. SVM : Support vector machines
SVR : Support vector Regression

SVM

1

Support vector machines

~~SVM~~ SVM : It is ~~nothing~~ nothing but called as support vector machines.

In ~~SVM~~ SVM, we can solve the both classification & Regression problem or algorithms.

- | | | |
|--------------------|-------------------------------------|-------------|
| (1) Classification | } → SVC (support vector classifier) | } intuition |
| (2) Regression | | |

1. Classification :

with respect to the classification, when we are solving, we basically say it as SVC (support vector classifier).

2. Regression :-

with respect to the Regression, we have ^{an} algorithms called, SVR (support vector Regression)

these are the 2 ~~algo~~ algorithms which we are using.

& we are trying to understand the 'Intuition' behind this entire algorithms.

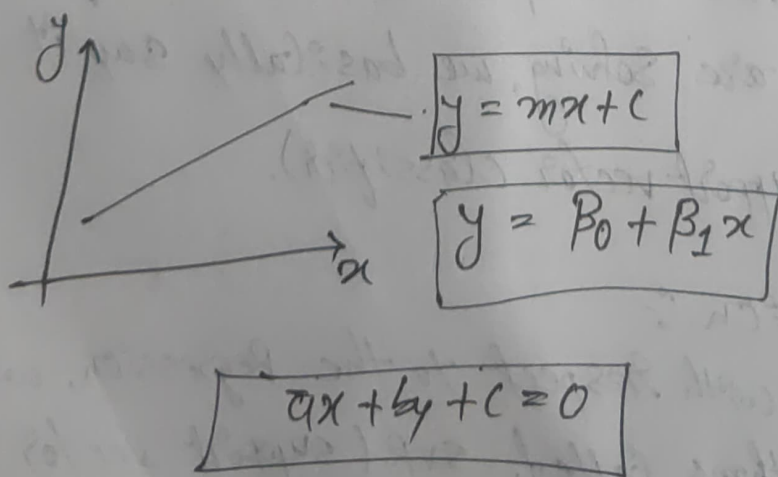
* This is the simple algorithms, here the basic things is required, linear algebra & all.

With respect to the 'linear algebra' there are something that we use to really need to know.

* what is the equation of the line (or straight line)?

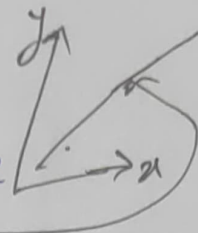
let's say,

I have this x-axis & y-axis.



~~$$y = -\frac{a}{b}x - \frac{c}{b}$$~~

$$y = -\frac{a}{b}x - \frac{c}{b}$$

So, I have straight line  is

basically nothing but $y = mx + c$

* Next, other thing we have discussed nothing but $y = \beta_0 + \beta_1 x$.

Can we write same equation as

$$ax + by + c = 0$$

this equation also a 'straight line' only.

I can convert this entire equation like this

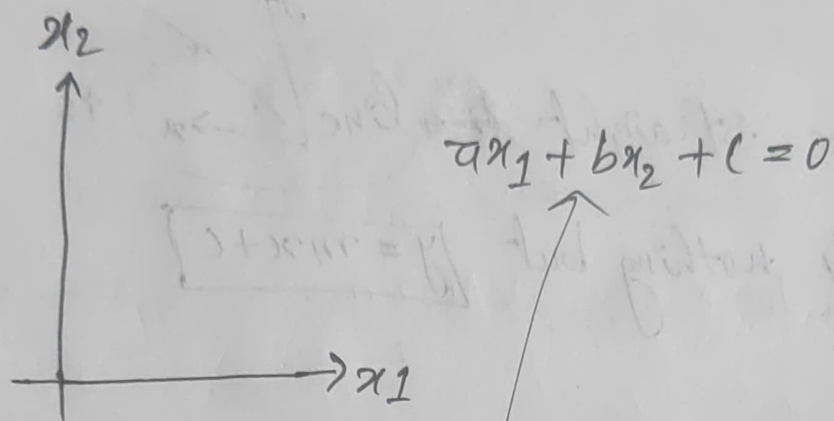
$$y = \boxed{-\frac{a}{b}}x + \boxed{-\frac{c}{b}}$$

↑
This is my Co-efficient

← This is my 'intercept'

* why I have written $\boxed{ax + by + c = 0}$?

If I have something like this



I have x_1 & x_2 , in this particular case my equation looks like this

$$\boxed{ax_1 + bx_2 + c = 0}$$

with respect to this, there is one notation, which we are also going to use, Can I write like this also,

$$\boxed{w_1x_1 + w_2x_2 + b = 0}$$

Same thing, just the notation is changed.

In short, we can write this equation as

$$w^T x + b = 0$$

w = weight vector

x = input vector

b = its a bias

T = Transpose

what this mean? w^T

I have $w_1 x_1 + w_2 x_2 + b = 0$

So, I have $w_1 x_1 + w_2 x_2$

over here

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \cdot \begin{bmatrix} x_1 & x_2 \end{bmatrix}$$

If I do the multiplication of this, or if I do a matrix multiplication, I will get the

$$w_1 x_1 + w_2 x_2$$

But, why I am writing $w^T x + b = 0$
 w^T is w Transpose

Because, this is the Transport of the

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \cdot [x_1, x_2]$$

this is how x_1 & x_2 Looks like

* at the end of the day, I am actually multiplying.

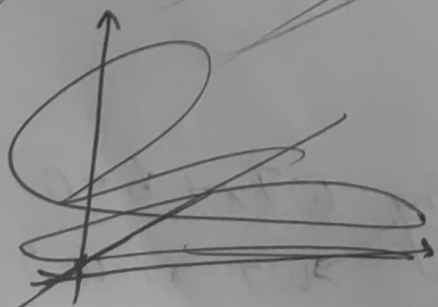
$$\cancel{w_1 w_2} \quad w_1 \cdot x_1 \quad \& \quad w_2 \cdot x_2$$

So, Can we basically write

$$wTx + b = 0$$

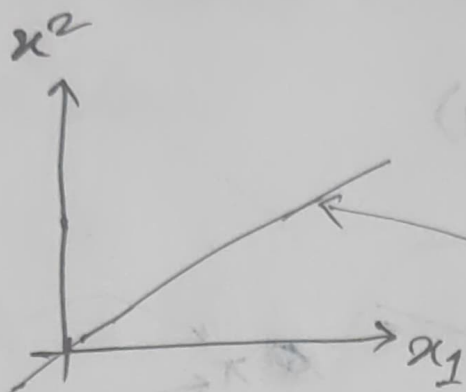
* Let's say,

this line passes through the origin.



* Let's say,

this line passes through the origin.



then what will 'b' value be? !

It will be zero (0) ~~to come~~

* So, Can we write 'w' Transport $x = 0$.

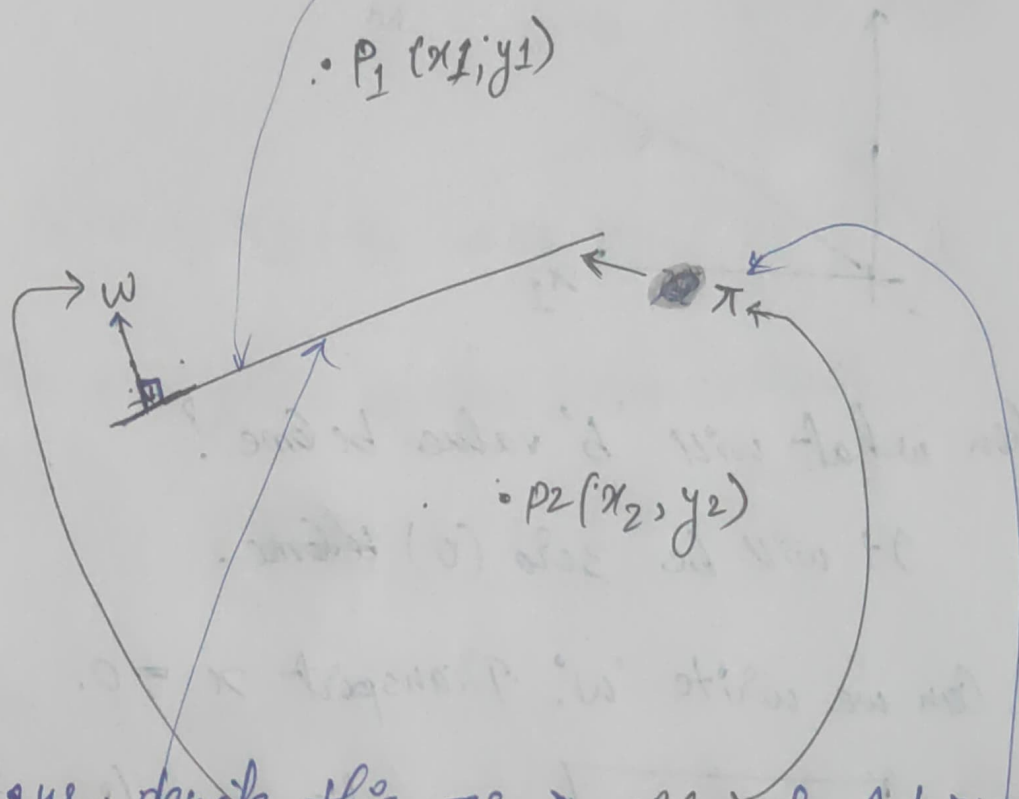
$w^T x = 0$ $\Rightarrow$ equation of a line passing through origin.

this is nothing but equation of a line passing through origin.

* So, this is what equation of line.

* let's say,

I have this line, it basically a 2D line.



let us denote this as a $p(\pi)$ & I have a vector, which is 'w'.

'w' is basically, ~~per~~ perpendicular to this line.

Let's say,

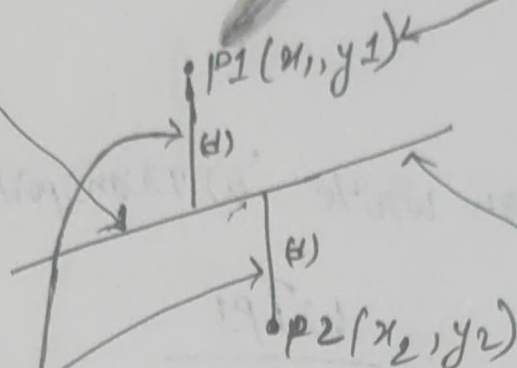
I have point - denoted by

$P_1(x_1, y_1)$. & I have another point $P_2(x_2, y_2)$.

same notation, same points.

over here this $\boxed{w}$ is a vector which is perpendicular to this plane.

* Now, in this scenario, if I really want to find the what is the distance of the point to this plane.



What is the distance of the point to this plane?

* In order to find thing, we really need to find out the distance of a point to the plane. (d) means distance.

* $P(\pi)$ is a line or plane.

* 2D : In case of 2D, this will a line.

* 3D : But, in case of 3D, I will get a plane.

like this $\rightarrow$

* Formula :

$$\frac{w \cdot T_{P1}}{\|w\|}$$

Now, this came from 'linear algebra.'

* If I really want to find out the distance of $(P1)$ to this particular plane, that is $P1(\pi)$.

then I can write 'w' transport (T) $\times$ $(P1)$ ^{multiply by}

$$\frac{w \cdot P1}{||w||}$$

divided by magnitude of 'w'.

This is a formula to basically find out the 'w' transport (T) multiply by $(P1)$ divided by magnitude of 'w'.

the region, we divide by magnitude of 'w'.

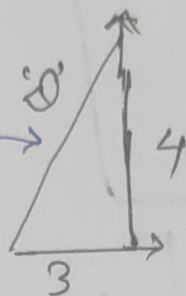
Because, we really need to get a 'unit vector.'

Question:- what is unit vector?

A vector which has a magnitude of 1, basically called as unit vector.

* $(3, 4)$

Let's say, I have a 2 vector which is $(3, 4)$.



this means, in the 'x' - the unit movement of 3. & I also have unit movement of 4.

If I ~~now~~ want to find out the distance between these 2 vectors.

Then, I just need to find out the this 'Hypothesis'.

In order to find out this hypothesis.

Then, I will write

$$\sqrt{9+16}$$

$$= \sqrt{25}$$

$$d = 5$$

Let's say, this distance (d) is my vector.

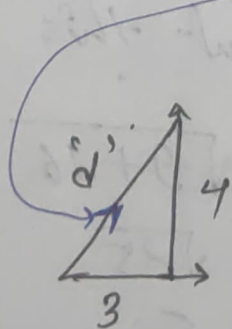
* If I really ~~was~~ need to convert this into a ~~unit~~ 'unit vector' which is written unit by this.

$$\hat{d} = \frac{d}{\|d\|}$$

I just need to divide 'd' by magnitude of $\|d\|$.

* what is this magnitude of 'd' $\|d\|$.

In short, what I am going to do, I really need to find out the unit vector $\hat{d}$, that basically means the direction will be same,



but the magnitude is changing, in order to do this.
I just divide $(3/5, 4/5)$

If we find out square root of this.

$$\sqrt{\frac{9}{25} + \frac{16}{25}}$$

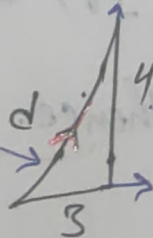
$$= \sqrt{1}$$

$$= 1 //$$

I am getting $\sqrt{1}$, at the end of the day,
I am getting $\boxed{1} //$

So, we can say this is my 'unit vector'.

To convert that to 'unit vector' I am basically dividing by the magnitude of this distance



$$\boxed{1/5}$$

So, this is nothing, but the magnitude,
we basically given this form as a equation.

* So, the region we are ~~divide~~ dividing this

~~$d = \frac{w^T p_1}{\|w\|}$~~

$$d = \frac{w^T p_1}{\|w\|}$$

Say the 'magnitude', $\|d\|$.

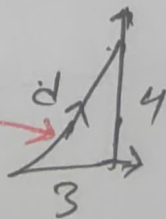
$\boxed{\hat{d}}$ ← this is how a 'unit vector' is specified. with this particular symbol $\hat{d}$

Question :

Q 1 How do Calculate the Hypothesis
distance?

why (9, 16) because this is

Hypothesis Hypothesis.



This is nothing but just square,
the 'pythagorean theorem', we have
discussed in statistics class.

Q.2. will unit vector always be 1?

yes, it has to be 1, it always be

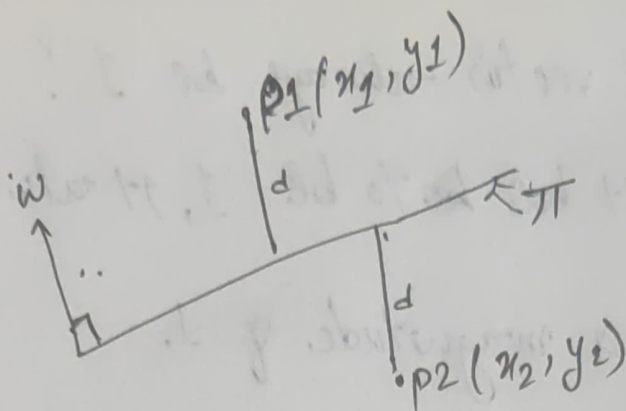
1.

It has a magnitude of 1.

So, in short unit vector is a way of
basically normalizing the data, so that we
are just focus on the direction.
we are not focus on the
magnitude.



*



$$d = \frac{w^T p_1}{\|w\|}$$

let's focus on this equation,

$$p_1(x_1, y_1)$$

* what is p_1 ?

it is a point, it is a 2 dimensional point, (x_1, y_1)

* what is $w^T p_1$?

the 'w' transport (T) multiply by p_1 , is nothing but, if I try to expand this equation,

this is nothing but, w mode P mode, it is

$$\|w\| \cdot \|p\| \cos \theta$$

$\cos \theta$ (theta).

this equation is there in the 'linear algebra.'

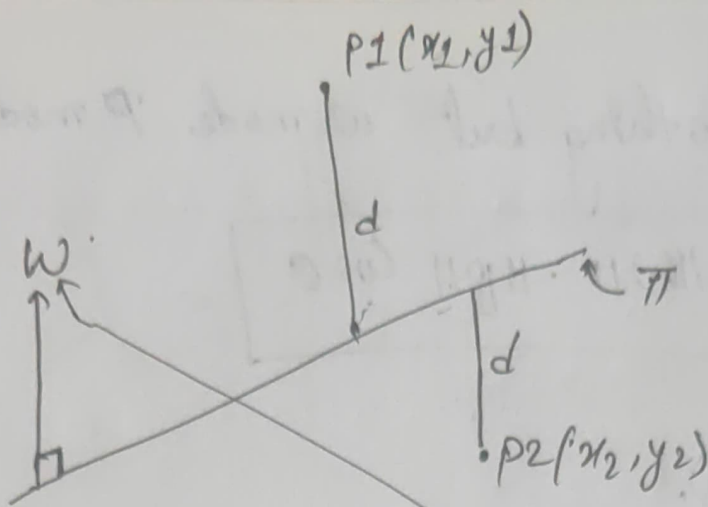
Again, we are not going to prove this

Q: If I really wanted to find out the distance between a specific plane.

this is nothing but that is

WTP1, here we can expand this equation by $\|w\| \cdot \|p\| \cos \theta$ mode of 'w' multiply by mode of 'p' multiply by $\cos \theta$.

Q: what do this formula actually convey?



over here with respect to this 'w'. A 'w' is a vector over here, like a co-efficient.

* what 'w' over here !

See this

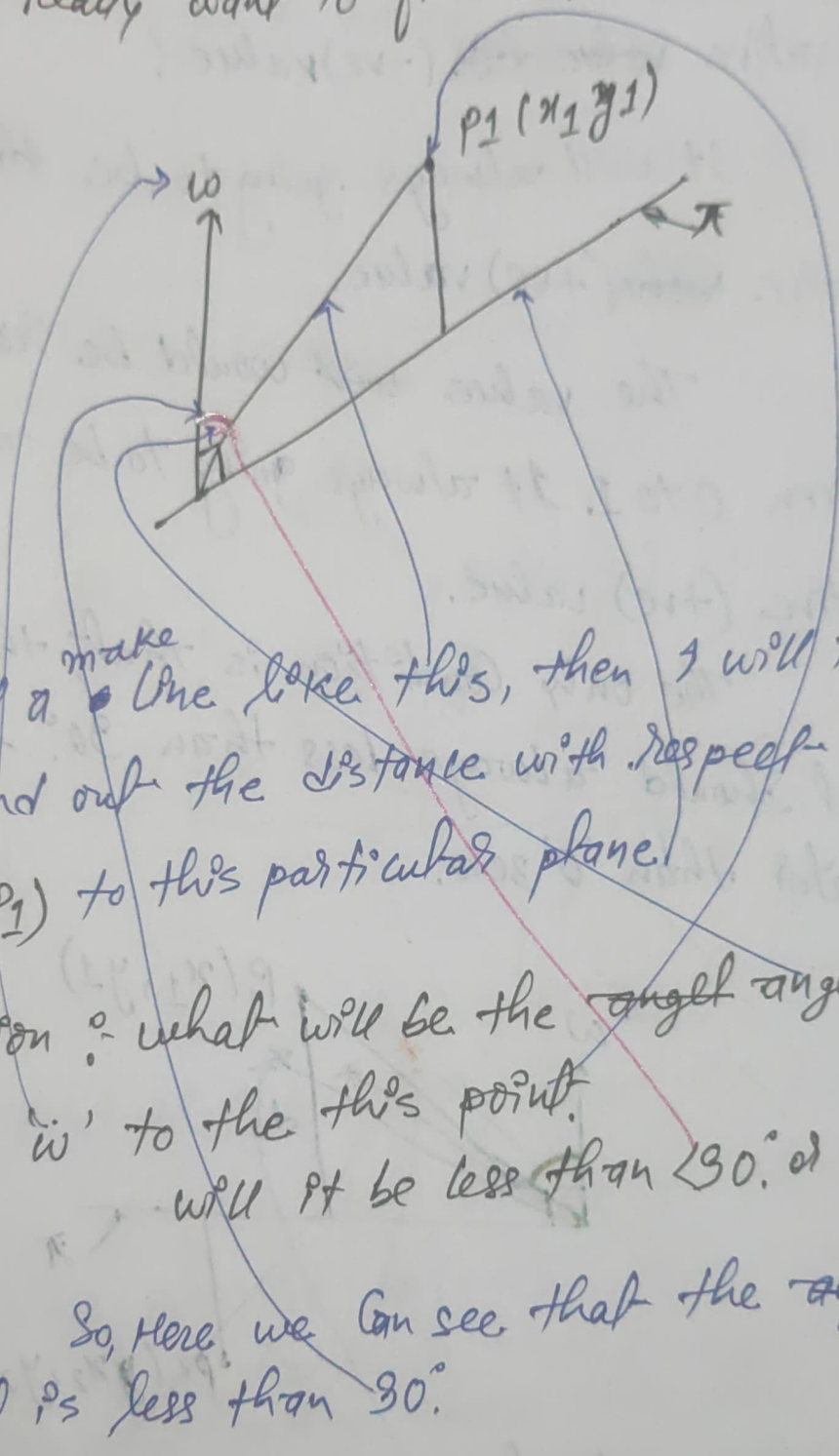
~~$$w_1 x_1 + w$$~~

$$a x_1 + b x_2 + c = 0$$

$$w_1 x_1 + w_2 x_2 + c = 0$$

'w' is nothing but vector of co-efficient,

* If I really want to find out the distance



I will make a line like this, then I will try to find out the distance with respect to P_1 to this particular plane.

* Question: what will be the angle, with this 'w' to the this point. will it be less than 30° or not?

So, here we can see that the angle is less than 30° .

Question: $\cos \theta = \frac{w \cdot P_1}{\|w\| \cdot \|P_1\|}$

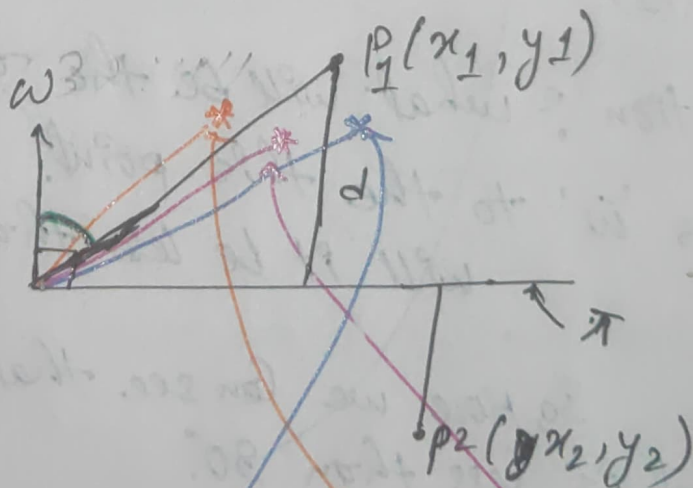
This value I am replacing in $\cos$, if it is between 0 to 30° , what value will I get out of it?

So, whether, it will be positive ^(+ve) value or a negative ~~value~~ (-ve) value?

It will always going to be ~~the~~ a positive ~~value~~ (+ve) value.

The value ~~will~~ would be range between 0 to 1. It always going to be a positive (+ve) value.

The only condition is that, the angle should always less than 90° & greater than '0' zero.



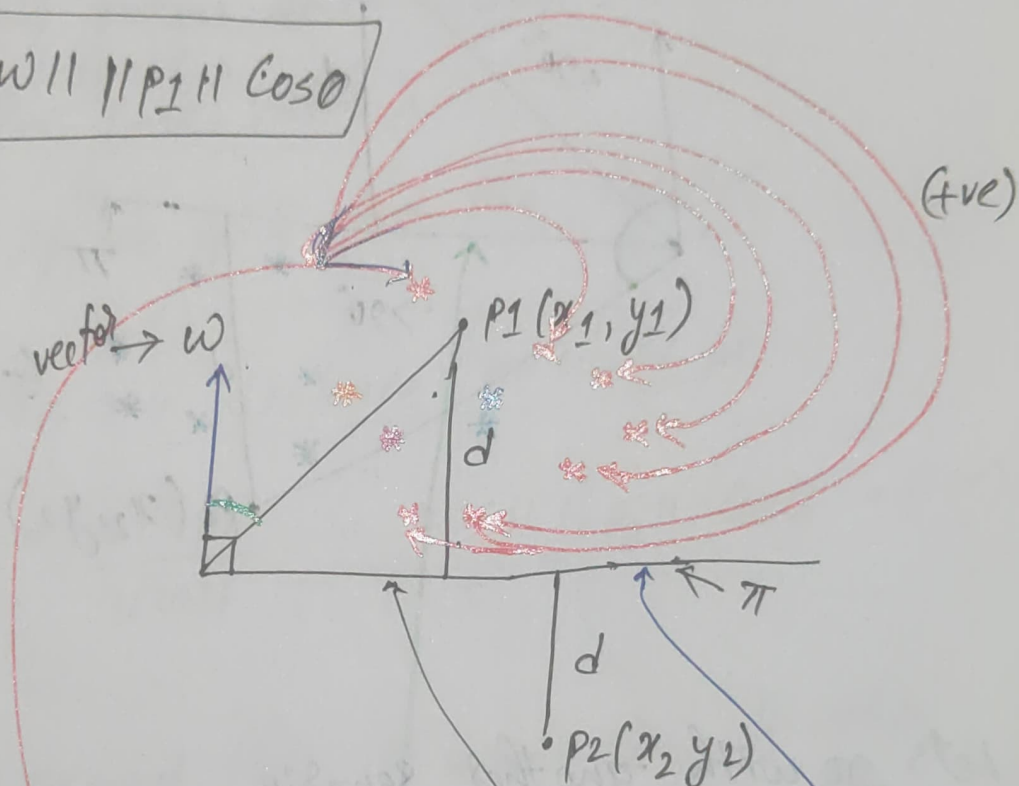
In this particular case, we will be able to see, with respect to this point, this point ~~of~~ this point.

when I create the angel, with
what will it be.

Yes the angle is always less than the 90° .

If I am trying to apply this formula

$$\|w\| \|P_1\| \cos \theta$$

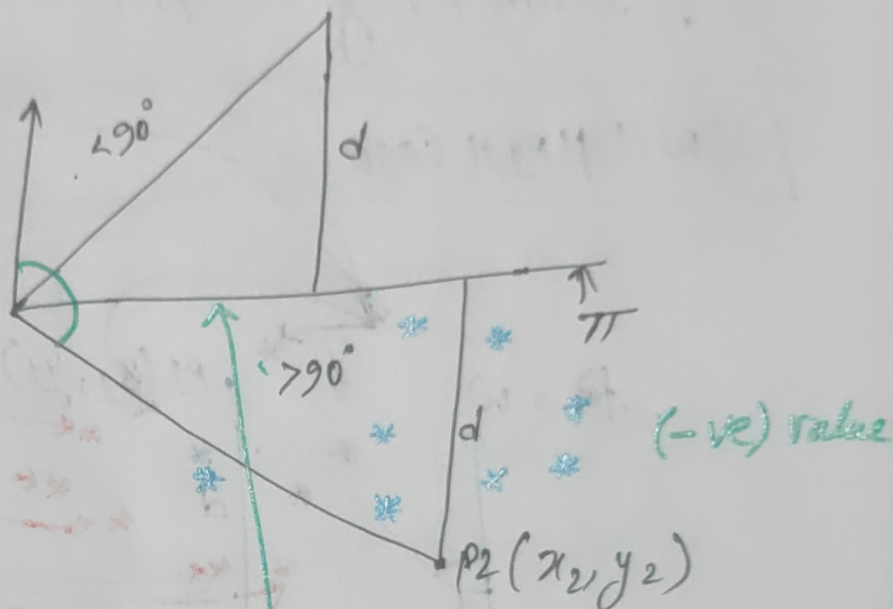


any point that is falling above this plane we can say distance to this plane, is always positive (+ve).

* Any point ^{that is} falling above this plane, if I try to calculate the distance, if I try to see the angle, it's always less 90° from this particular vector.

It's always going to be (+ve) in short. $\cos(0-90)$ always (+ve) positive number.

2nd Example:



Let's go with another scenario.

For this (P_2) if I draw a angle,

* what is the angle, I am getting. will it be ~~ge~~ greater than 90° or not?

So, in this scenario, it will be ~~ge~~ greater than 90° . ~~Can~~

Can we conclude, every point, which is below then this plane.

If we ~~find~~ try to find out the angle, it is always going to be greater than 90° .

If it is ~~also~~ greater than 90° , basically means, ~~it~~ it always show me a negative (-ve) value.

* If we go to find the $\cos 90^\circ$,

For this my distance ~~is~~ would be

$$d = \frac{w \cdot T \cdot p_2}{\|w\|} = \|w\| \|p_2\| \cos \theta = (\text{ve})$$

If we expand this, it would be θ this θ value always greater than 90° , so I am going to get the value of (-ve) negative.

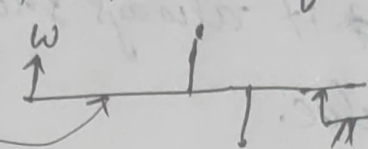
* Note:

Go to find the $\cos 91^\circ$, $\cos 95^\circ$, $\cos 120^\circ$, $\cos 140^\circ$.

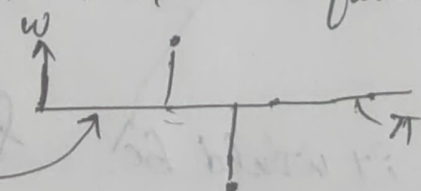
If we go to find out the all of these value will be (-ve) negative.

* Conclusion:-

The conclusion is that,

- * Any point that are falling above the plane  & if we try

to find out the distance, it will always going to come to the positive (+ve).

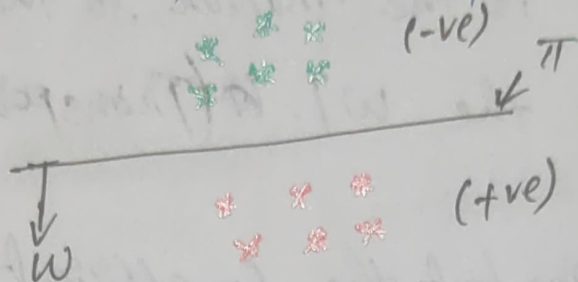
- * Any point that are falling below this plane  & we try to

find out the distance, & which has a angle with respect to the vector, then over here we will be able to find out the distance negative (-ve) distance.

* Question :-

Let's say,

This is my plane π .



& this time my vector is facing downwards 'w'. But it is perpendicular.

* The above point distance would be positive or negative?

This case & my all the point distance would be (-ve) negative.

* The below points would be (+ve) positive distance

* Question :-

what is $P_1(x_1, y_1)$

If I draw 2 dimension point

It will have 2 values,

with respect x_1 & y_1 , the x -axis & y -axis



will be there.

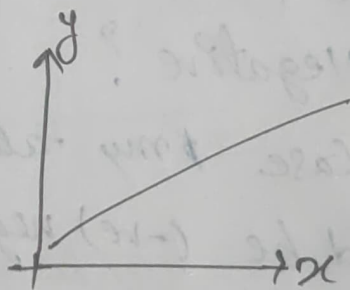
Question :-

what is the matrix, so that we can calculate the WT. (Transport).

'w' is nothing but the Co-efficient.

Let's say,

I have 2 dimension point. ~~(x, y)~~.



$$y = mx + c$$

If I have (x, y) ~~then~~ then how many Co-efficient, I will be having to create best fit line?

the equation is simple $y = mx + c$

So, I will have 1 Co-efficient.

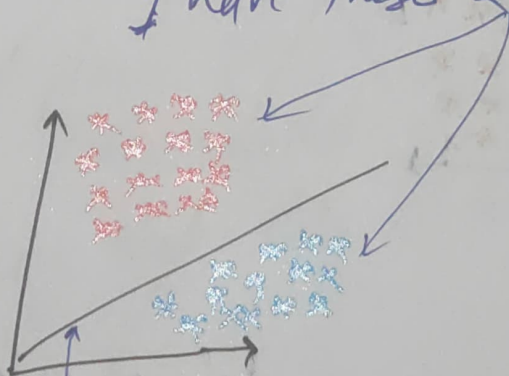
Geometric Intuition behind support vector machine?

(1) Logistic Regression :-

~~with respect~~
SVM is a algorithms.

with respect to the 'logistic regression', the main behind is that,

Let's say, I have these 2 data points,



Logistic regression

'logistic regression' mostly used for 'binary classification'.

End of the day, Geometric intuition, was that, we really need to find out the 'best fit line', which will be able to classify this particular points.

This was with respect to logistic regression.

(2) Support vector classifier

This support vector classifier (SVC) also does the same task, it can also do with 'binary' any number of categories.

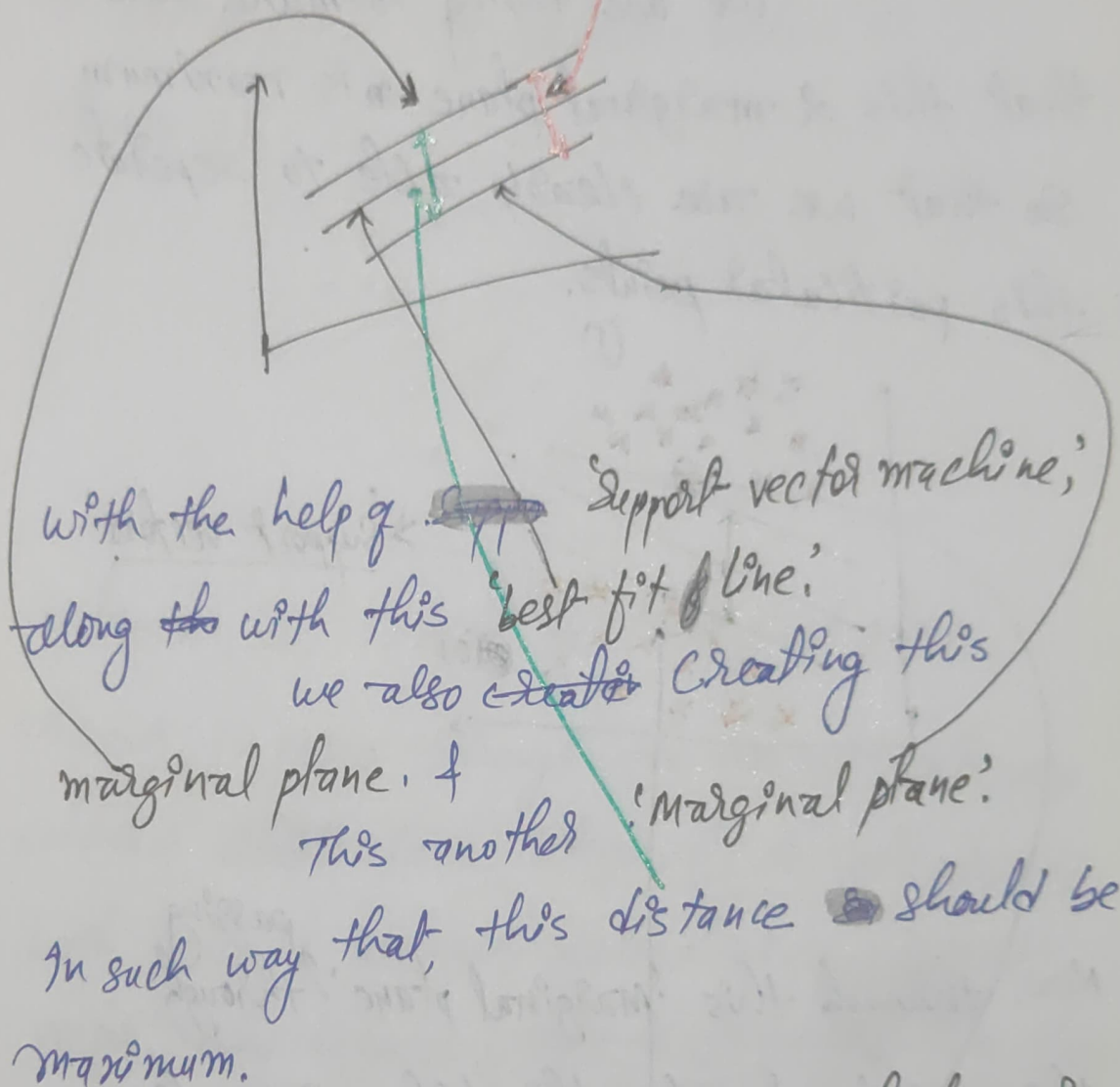
Let's say,
I have 2 categories.



Now, in (SVC) also, we try to find out the 'best fit line', along with this 'best fit line', we also try to calculate the 2 marginal planes

That will be passing through this one of the nearest point of the other category & it will be passing through the other point nearest through this categories also.

such that this distance should be ~~maximum~~ maximum.



with the help of ~~Support~~ 'Support vector machine,'
along ~~to~~ with this 'best fit line,'
we also ~~create~~ creating this
marginal plane. &
This another 'marginal plane'.

In such way that, this distance ~~is~~ should be
maximum.

Ques Question: what is 'Marginal plane'?

It is a another line, just like a
'best fit line,' which passes through the nearest
point over here.

So, over here what, we are doing with ~~SVM~~,
SVC (support vector classifier) is that, along
with this 'best fit line'.

we are ~~also~~ also creating the ~~2nd~~ 2 'marginal plane.'

we are trying to make sure that, this 2 marginal plane is maximum, so that we are clearly able to separate this particular points.

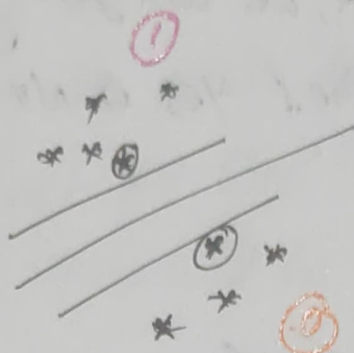


Now, ~~through~~ this 'marginal plane' ^{passing} through this points & also this below one also passing through this particular ~~on~~ point.

This points are basically called as 'Support vectors'.

* Here, we are understanding the Geometric Intuition, we have not ~~discussing~~ discussed mathematic concept yet.

* So, let's assume. ...above 'straight line' is
a Category (1) & 'straight line' below as a
Category (0).



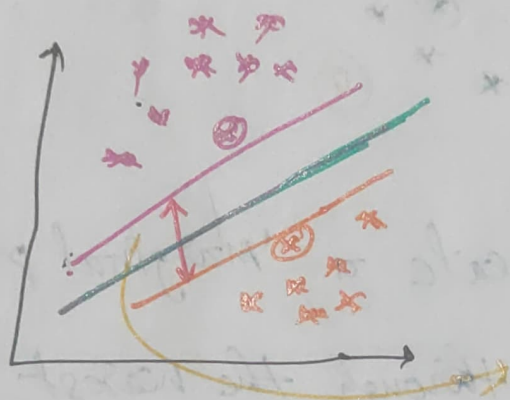
whenever, we create a 'marginal plane.'
whenever, it passes through the nearest Category
whether ~~above~~ above Category (1) or Category (0)
both, this is actually called as a 'support
vector.' why because ~~this is helping~~ ~~these~~ ~~these~~
~~points are~~
points are helping us to create this particular
'marginal plane.'

Similar with respect to the other
points near to 'margin plane,' we can call
it as 'Support vector.'

we can have more than 1 support
'1 support vector' also.

Question:

In the real world would some scenario, do we think, we will getting so clearly clearly seperable line? yes or no?



Hard margin

No, we will not be getting because there will be so many points that will be overlapping.

If we get this clearly seperable lines, then this is basically called as 'Hard margin'.

* Hard Margin :-

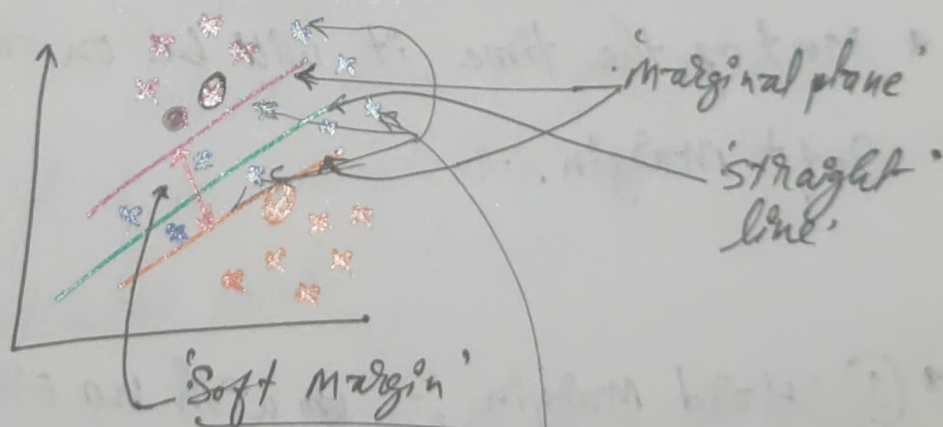
~~It is possible~~

In Hard margin, we do not find any errors & we will be able to clearly separate all the points by using the 'Marginal plane' & 'best fit line'.

There will be no error at all.

* Soft margin :-

Let's say, in the real world scenario, there



would be a lot of overlapping. Some points will come here & there like this, they will go in Category (1) & also go in Category (0) & some go in between these 2 categories.

In this case, whenever we are trying to find out the 'Marginal plane' along with

best fit line.  with the same errors.

we basically say this as a 'Soft Margin.'

In real world scenario definitely we will be ~~not not~~ not getting clearly separable, there will be some errors.

In this particular scenario, we say it as 'Soft Margin.'

* Most of the time, it will be on a Soft Margin.

* (1) Hard Margin :-  with no errors it's a Hard Margin.

* (2) Soft margin :- with the errors it's a Soft Margin.

* If we compare these 2 margin A & B?
which one will be good margin?



If we look into both margin; we can see that the (B) margin is better one.

Because these margin plane is very high, & we are clearly able to separate the particular point.

* My 'SVM' model should be able find part (B) kind of model 'margin plane' even though we can get a part (A) kind of margin plane also from the same data points. Distance should be maximum, what we try to find it over here. in this marginal plane

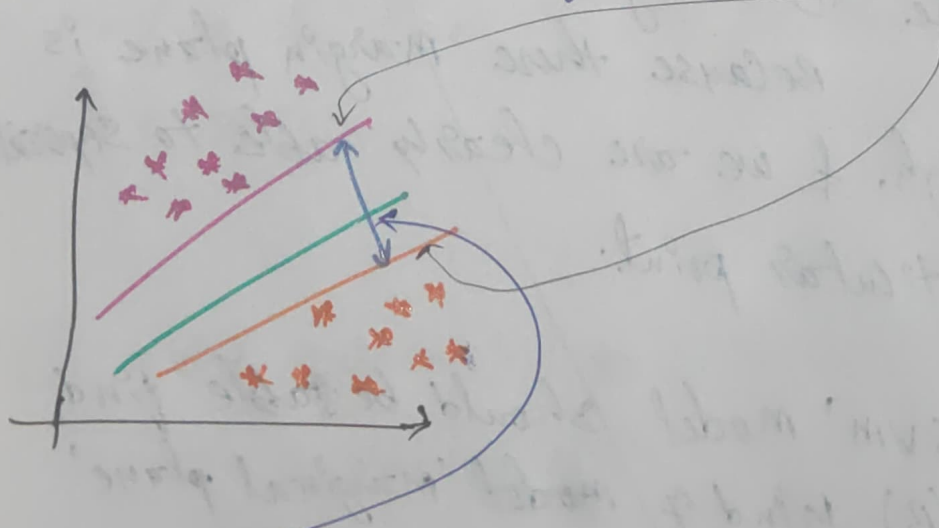
* Mathematical Intuition ~~is~~ behind the SVM (Support vector machine) ?

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Let's solve this,

What is our main aim over here ?

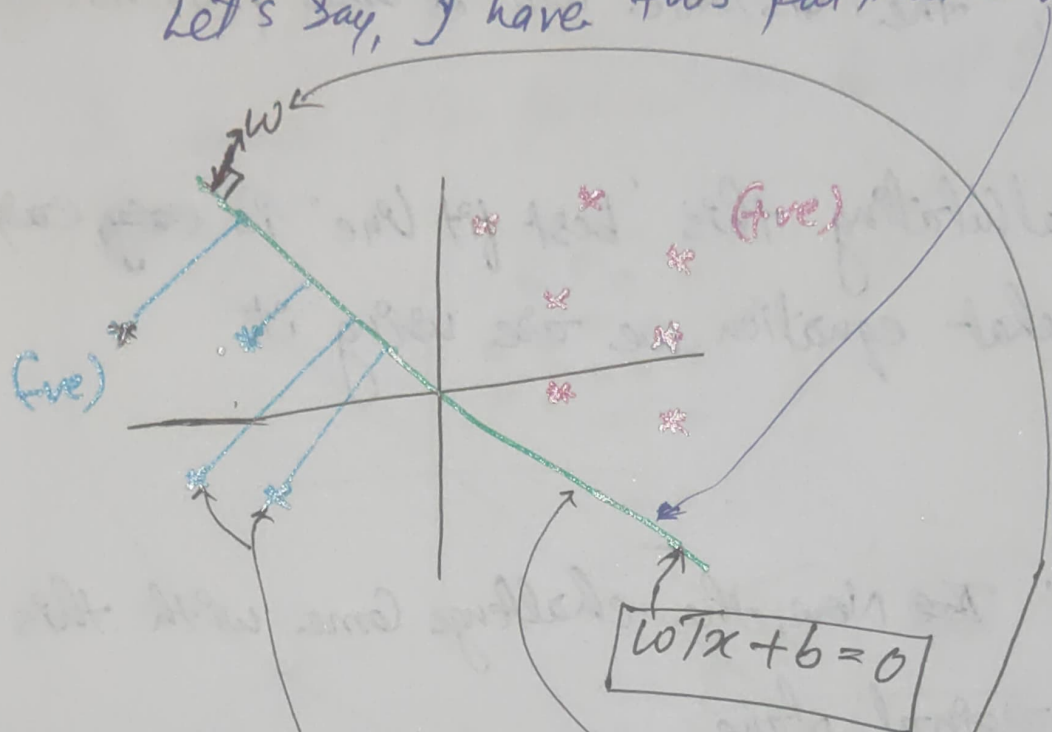
To find out the 'best fit line', along with that we really need to find out the margin plan.



One more thing is that, this 'margin plan' should be equidistant (समान दूरी) from this particular 'best fit line'.

Let's solve this

Let's say, I have this particular line,



Q:- what is the equation of this line?

A $\boxed{w^T x + b = 0}$ & this is my 'w' vector.

* If I have any point below this, the distance between this particular point with plane would be positive (+ve) or Negative (-ve)?

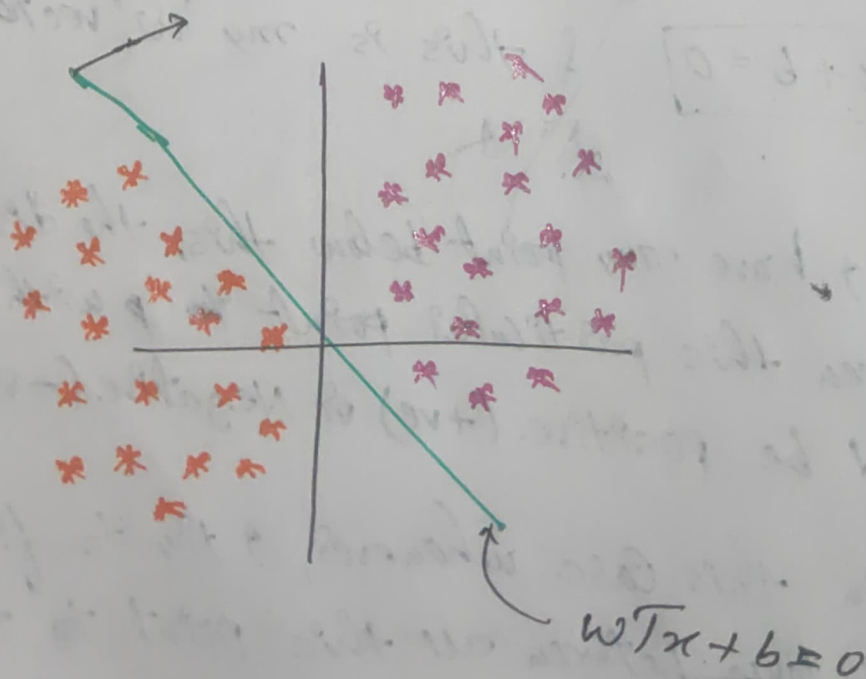
In this case whenever, I try to find out the distance between all this point to this plane, it always going to come it as Negative (-ve).
the distance basically Negative (-ve).

Similarly, whatever points are coming above this, the distance will be come positive (+ve).

* Calculating this 'best fit line' is ~~easy~~ easy, what equation we are using it.

* Q: ~~The~~ Now, the challenge come with this marginal plane.

Let's draw points *

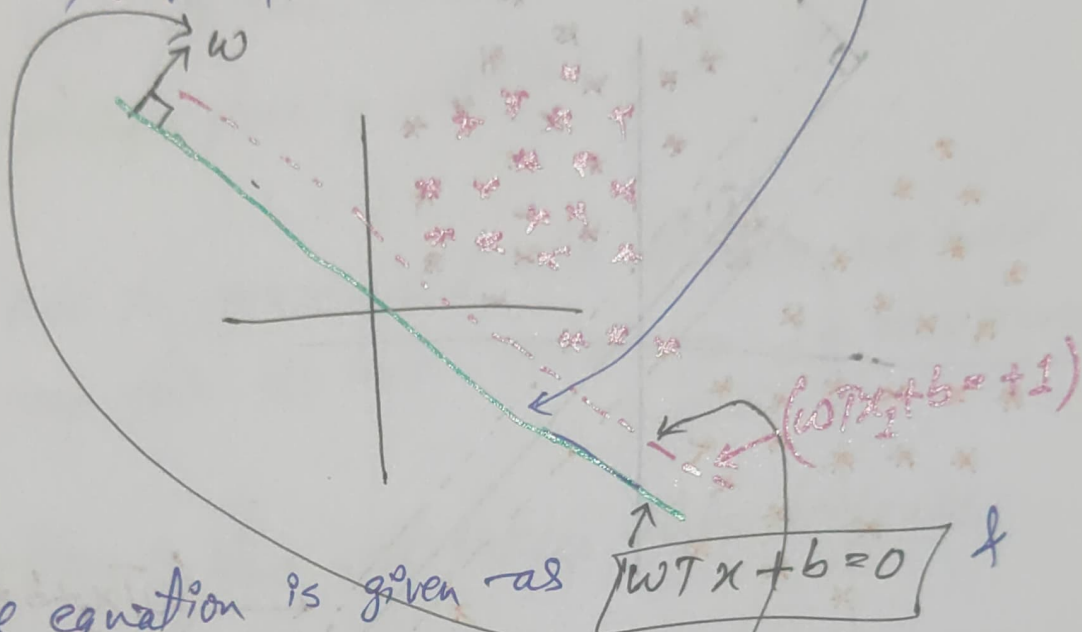


I have 2 Category data points over here,

~~Hard margin.~~

Hard margin. & Soft margin.

So, obviously, I can find out my 'best fit line'



The equation is given as $wTx + b = 0$ &

above in this particular plane, if 'w' is in this direction.

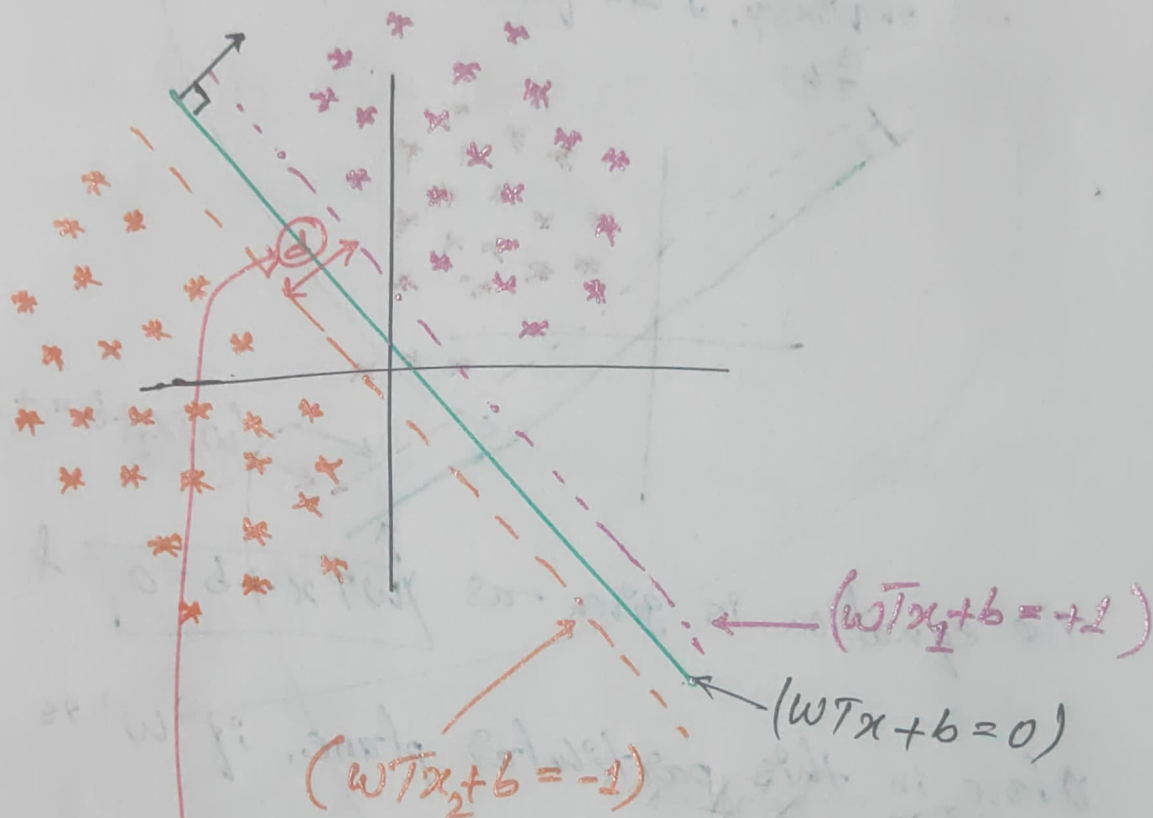
all the ~~distance~~ point above this ~~plane~~ plane, the distance will be ~~positive~~ positive (+ve).

Now, Can we create a 'best fit line' 'Margin plane', which passes through this & provide ~~simple~~ simple equation $wTx + b = +1$

Can I give this as a (+1) over here.

* Similarly,

I will try to create another margin plane.



which will be below then this particular plane, & this is given as $(wTx_2 + b = -1)$

* what was our main aim?

our main aim was to make this distance (d) maximum. $d = \text{maximum}$

So, we are getting 3 simple equation.

① Margin plane $= wTx_1 + b = +1$

② Best fit line $= wTx + b = 0$

③ Margin plane $= wTx_2 + b = -1$

In order to maximize this distance, we need to keep this distance maximum $d = \text{maximum}$.

* Question : How will I keep this distance maximum?

~~The equation is~~

Let's ~~test~~ ~~some~~ calculate this

$$wTx_1 + b = +1$$

$$wTx_2 + b = -1$$

$$\begin{array}{ccc} (-) & (-) & (+) \end{array}$$

$$\hline wT(x_1 - x_2) = 2$$

Now, main thing is I need to keep this particular

(d) distance ~~very~~ ~~high~~ very high.

The space should be high as possible,
what we will do then.

If I will do the subtraction
on both of this equation.

Cost function

$$w^T x_1 + b = +1$$

$$w^T x_2 + b = -1$$

$$(-) \quad (-) \quad (+)$$

$$w^T (x_1 - x_2) = 2$$

Can, I write, ' w^T ' with respect to $(x_1 - x_2)$ this
will entirely ~~deducted~~ & it will be ~~be~~ 2
dedicated

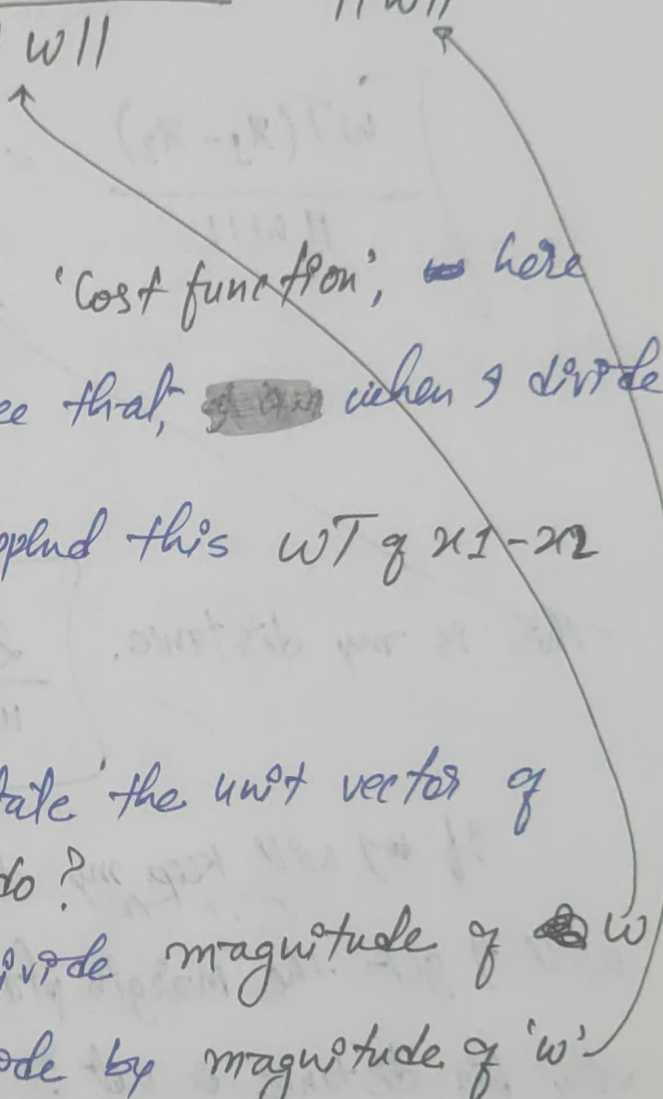
In short; we are calculating the
Distance (d) between these 2 margin plane.

* why we are doing this?

Because, I need to derived my
'Cost function'.

my 'cost function' is very important over here.

*

$$\frac{WT(x_1 - x_2)}{\|w\|} = \frac{2}{\|w\|}$$


with respect to this 'cost function', ~~we~~ here we will be able to see that, ~~if we~~ when I divide this, what will happen this WT of $x_1 - x_2$ ~~will~~ divide by 2.

In order to calculate the unit vector of 'w', what we will do?

Can we divide magnitude of ~~w~~ w

& here also divide by magnitude of 'w'

* we are trying to find out the difference between ~~margin~~ these 2 margin plane, where in the distance should be maximum.

Now, in order to convert this unit vector, because we don't have to worry about the magnitude ($\| \cdot \|$), because we ~~we~~ are

dividing by magnitude of 'w' $\|w\|$

& this is what we are finally getting.

$$\left[\frac{wT(x_1 - x_2)}{\|w\|} = \frac{2}{\|w\|} \right]$$

This is my distance.

$$\frac{2}{\|w\|}$$

If I will keep my distance maximum,
will I get the 'margin plane' distance a
very big distance or not?

* what is our main aim?

Now, I am going to writing the 'cost function'.

Cost function

~~Maximum~~

Maximize

$$\frac{2}{||w||}$$

← distance between
marginal planes

w, b

my 'cost function' is nothing but, I need to maximize, what value $\frac{2}{||w||}$ by changing the values of (w, b) why?

Because, I have the w value & b values.

$$wT x + b = +1$$

I have to keep on changing this (w, b) , in such a way that, I need to get this 'margin plane', such that the distance between them should be maximum.

This is nothing but distance between margin planes.

* what I did?

I try to find out the difference between them & I get this as my difference $\left(\frac{2}{||w||}\right)$

Now, my aim is that, I need to maximize
 $\left(\frac{2}{\|w\|} \right)$ by changing the (w, b) &
this is what I am actually getting, with
respect to distance between marginal planes.

* public question

By changing (w, b) . mean?

If mean change it, we need to continuously
iterated.

~~the~~ similar to that, we did for

(β_0, β_1) weight updation formula

How we did for 'Gradient descent'

Similarly we have to do ~~this~~ for this

(w, b) , so that the particular $\left[\frac{2}{\|w\|} \right]$ value

that we get should be highest.

* So, finally we got this 'Cost function' which we really need to work on.

$$\text{maximize}_{(w, b)} \frac{2}{\|w\|}$$

* Along with this, why we divided by 'mode'?

The reason is that, we divided by 'mode'.

$$\frac{w^T(x_1 - x_2)}{\|w\|} = \frac{2}{\|w\|}$$

(unit vectors)

Here, we need ~~to get~~ 'unit vectors', with respect to this, we need to get the 'unit vectors'.

If we want to get the 'unit vector', we have to divide by 'mode' of $\|w\|$ or 'w'.

* So, ~~the~~ we saw previously what is unit vector? The unit vector is a 'magnitude' $= 1$. (M = 1)

* Constraint

So, now along with this 'cost function', what, I am also doing is that, I am going to add some constraint.

(1) Constraint such that $y^i \in \begin{cases} 1 \\ -1 \end{cases}$

$\cdot wTx + b \geq 1$

$\cdot wTx + b \leq -1$

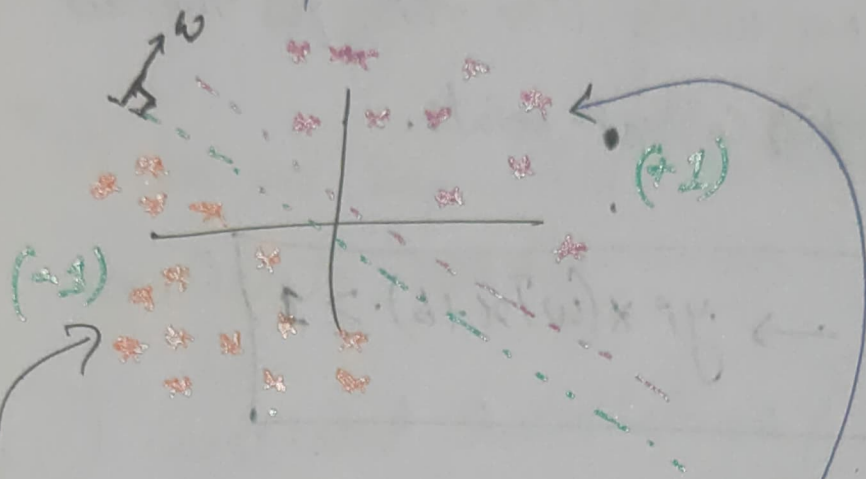
y^i or $y(i)$ is usually my true output, with respect to the dataset.

*(1.) This true output, if I say my $[wTx + b \geq 1]$ or $wTx + b$ is greater greater($\geq$) than equal to 1.

what is my correct predicted output for this?!

Can we write (+1)

that basically means, I am coming to this particular points. $(+1)$ & (-1)



So, whenever, my $(wTx + b \geq +1)$ $wTx + b$ is greater than or equal to $(+1)$, that basically I am ~~saying~~ saying that this is pointing to this particular point & this particular point is label is 1.

* So, whenever my $(wTx + b \geq 1)$, I am going to get the output as 1.

*(2) If my $(wTx + b \leq -1)$ or $wTx + b$ is less than -1 .

This scenario, my value would be (-1) .

* Can we say, this is a condition for all correct classified points.

* Now, considering the 'constraint'.

~~Can we write like this, for the all~~
~~classified~~ correct points.

$$(2) \text{ Constraints } \rightarrow y_i \times (w^T x + b) \geq 1$$

~~If I multiply~~

Constraints, simply like this, if
I multiply $y_i \times (x)$ multiply by $(w^T \text{transfer } x + b)$

I am always going to get ~~the~~ this as ~~greater~~
greater ($\geq$) or equal to $\boxed{1}$. all the correct
classified points.

Just multiply, this

* ① 1 multiply by $-1 = \geq 1$ or greater
than equal
to 1.

* ② -1 multiply by $-1 = \geq 1$ or greater
than equal to
(1)

* (-1) multiply by (-1) always going to get it as ≥ 1 or greater than equal to $\boxed{1}$.

* So, can we say this both the constraints are same.

$$\textcircled{1} \text{ Constraint such that } y_i \begin{cases} 1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq -1 \end{cases}$$

$$\textcircled{2} \text{ Constraints } \rightarrow y_i \times (w^T x + b) \geq 1$$

I have just simplified this constraints that ~~casually~~ means this is my truth point, when I multiply $y_i \times (w^T x + b)$, I am always going to get as $(+1)$. & this is for both the Negative (-ve) point & for the positive (+ve) points.

* Now,

Let's change a little bit, because these are equations. Let's play with the equation.

Initially we wrote the

$$\begin{array}{ccccc} \text{maximize} & \frac{2}{\|w\|^2} & \rightarrow & \text{Minimize} & \frac{\|w\|^2}{2} \rightarrow \text{loss function} \\ w, b & & & w, b & \downarrow \\ & & & & \text{Minimized} \end{array}$$

① Maximize $\frac{2}{\|w\|^2}$
 w, b

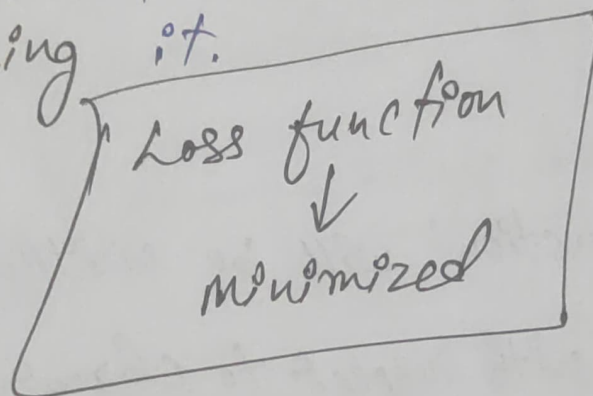
This is my 'cost function', I have to maximize this value by changing (w, b) .

② Minimize $\frac{\|w\|^2}{2}$
 (w, b)

Can I write, same equation, like this, where I need to minimize $\frac{\|w\|^2}{2}$ or magnitude of w . by 2, by changing (w, b)

Here, my 'cost function', I have written on left hand side to 'maximized'. &

at the end of the day, my every ~~loss~~ loss & cost function that we have learn till now, we need to focus on minimizing it.



that what we have learn from the 'Linear Regression' & the 'logistic Regression' we really need to minimize it.

So, we can write like this

$$\left[\begin{array}{l} \text{minimize} \\ w, b \end{array} \quad \frac{\|w\|}{2} \right]$$

inverse of that.

* Cost function

$$\min_{w, b} \frac{\|w\|}{2} + C \sum_{i=1}^n \xi_i \quad \xrightarrow{\text{Soft Margin}} \text{Hinge loss}$$

my 'cost function' will be written as
 minimized with respect to changing (w, b) ,
~~magnitude~~ magnitude $\|w\|$ of 'w' by 2.

$$\min_{w, b} \frac{\|w\|}{2}$$

obviously, we have written constraints over there.

$$+ \sum_{i=1}^n C_i \xi_i$$

~~So, we will going to add~~
 This is apply for 'Soft Margin'

• & this is super important.

Because end of the day, we will not get a problem statement, that is exactly a hard margin.

So, we will going to add, *2 important parameters.

$$\boxed{C_i \sum_{i=1}^n \xi_i} \longrightarrow \underline{\text{Hinge Loss}}$$

• 1 is C_i , submission (ξ_i) of $i=1$ to n & I am going to mention this as (ξ_i) or Eta of i .

* So, this entire this, when I am adding this become something called as Hinge Loss.

* we have heard or learn previously like Huber loss.

This is also a one new loss function, which is called as 'Hinge loss'.

* what is the meaning of this

$$C_i = \sum_{j=1}^n \xi_{ij}$$

~~① C_i or C_j~~

~~C_j is nothing but, how many points, we can ignore for misclassification.~~

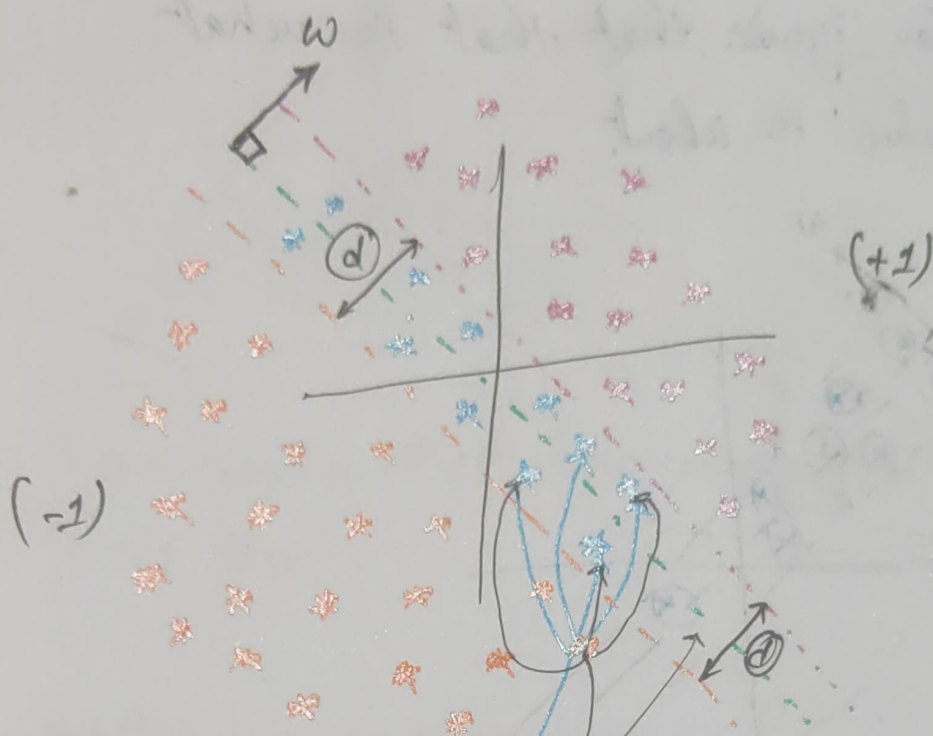
Hyperparameter
↑ its function
is

① C_j or C_i

C_j is nothing but, Hyperparameter
↑ its function is nothing but, how many
points, we can ignore for misclassification.

what this basically means!

* Let's consider in this way.



Over here, Can we say this is completely 'Hard margin'?

In a realworld scenario, will we get this kind of points, 'No right'.

If I am having some points like this overlapped both sides,

So, what does 'c' means!

even though, if I create those 'margin plane', I am saying that, fine, let's some of the points being wrong.

How many points are coming wrong?

we can ignore that, that is what the 'c' value is about.



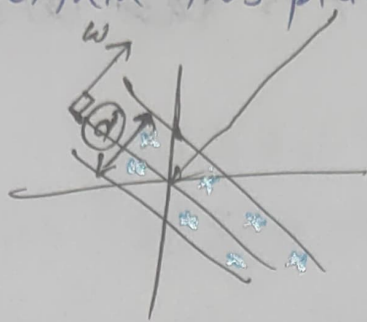
How many points are wrong over here?

we can see lets ~~say~~ 7 points over here.

So right now, for this particular problem statement, I have taken the 'c' value as an 7.

So, this is specifically a 'Hyperparameter', just saying that, how many point, even though, if there is a ~~mistake~~ mistake

we are ignoring them.
within this plane.

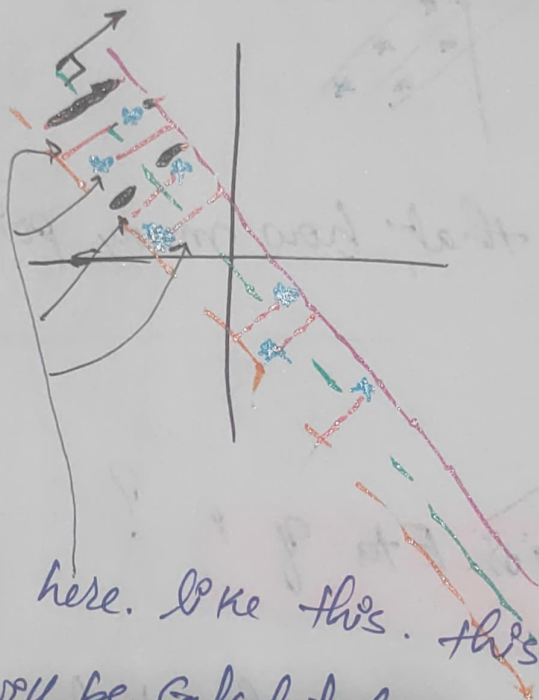


I am saying that, how many points, we just ignore that.

* $\ell(\xi_i; i)$ or Eta of i ?

ξ_i or Eta is nothing but, this is another hyper parameter; it is nothing but the ~~Sum~~ ~~Subm~~ Summation of the distance of the incorrect data points, from the marginal plane.

Along with this 'C & P', we are taking the distance also. It is ~~sub~~ calculating



from here to here. like this. this ~~sub~~ summation will be calculated.

* ~~these~~ So, this both combined together

$$C_i \sum_{i=1}^n f_i + E_{\text{margin}}$$

If when we do the ~~sub~~ submission of the distance, these are ~~2 type~~ 2 hyper parameters, we are going to use to create this 'best fit line' along with the 'margin plane.'

This entire ~~thing~~ things is called as 'Hinge Loss'

It is a 1 hyperparameter,

* C of ℓ :- It says that, how many ~~par~~ parameters we can ignore from the misclassification

* ξ_{ℓ} of ℓ :- It is nothing, but ^{this entire} ~~Submission~~ distance of the incorrect data points from the marginal plane.

* Public Question :-

Why do we desicrate the fraction?

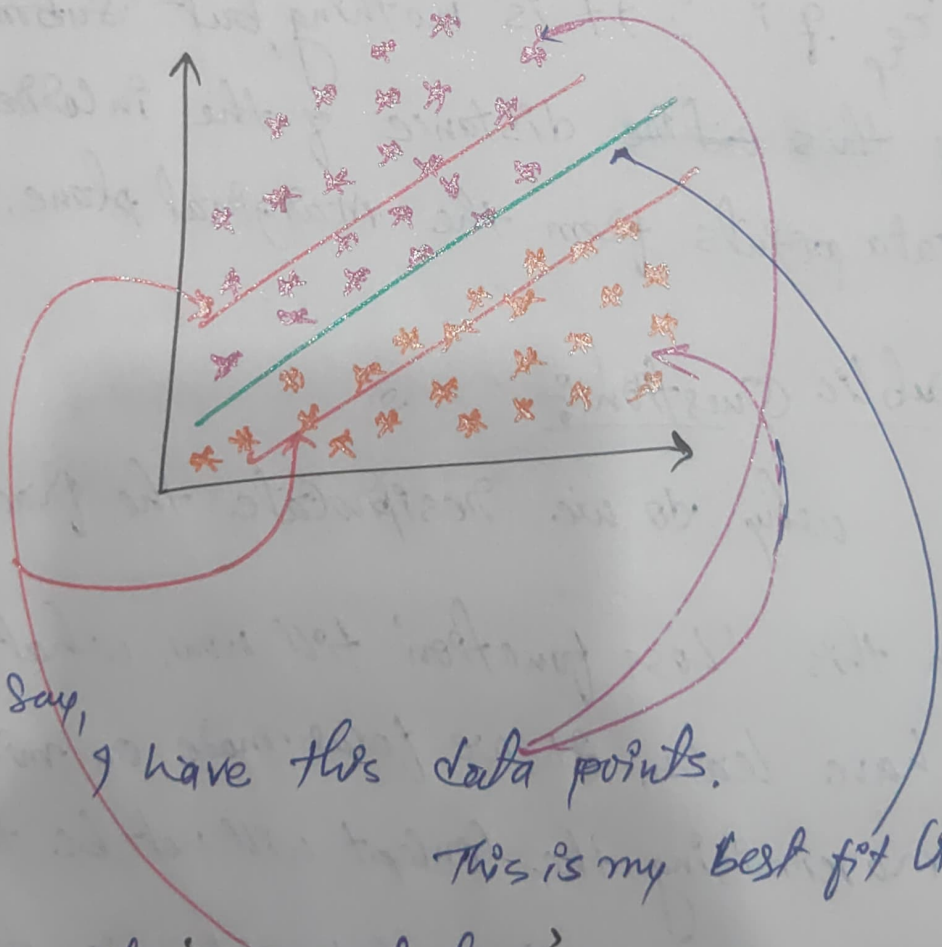
In this 'Loss function' till now, whatever we have learn, ~~at~~ we focus more on minimizing, a maximizing the concept will not be there.

We really need to minimize those loss, by finding out the derivative of all.

So, these all will be find out the derivative in a efficient way.

* Question :- public Question
what is this ξ_{margin} ?

ξ_{margin} or ξ is nothing but submission of incorrect data points from the ~~margin~~ marginal plane.



Let's say, I have this data points.

This is my best fit line.

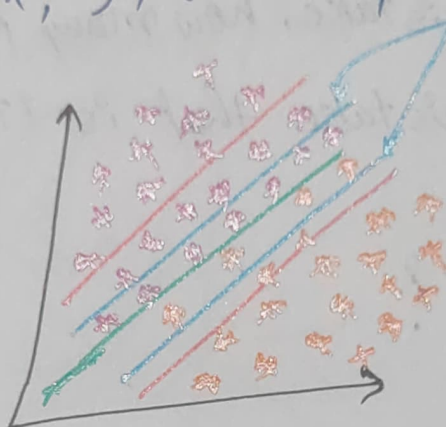
This is the 'marginal plane'.

In this scenario my 'C' value will be high or low?

It will be higher because so many data points are there.

* If I will restrict my ' ϵ ' value, what will happen to this marginal plane?

Then, it will keep reducing like this



* If I will restrict my ' ϵ ' or ' η ' value? then this marginal ~~is~~ plane, will keep reducing.

That is how it play a role to finding this.

* C_i & C_j ?

This is basically means how many points, we can ignore for misclassification.

In the 'soft margin' even though we are making mistake, how many points we can make a mistake that is it.

* ξ_i & η ?

Along with that we are also making sure that the submission of all the distance of the incorrect data points from the marginal plane.

~~* we are using 2 things~~

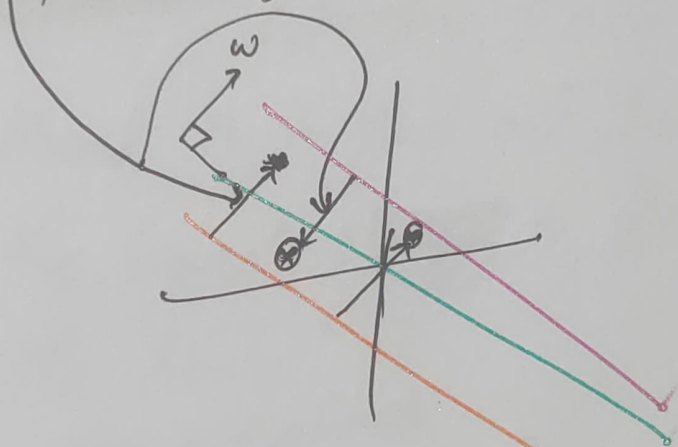
~~1st of all how many points can be incorrectly classified.~~

~~Then we also try to find out the distance between the this particular points to~~

* we are using 2 things.

1st of all how many points can be incorrectly classified.

then we also try to find out the distance between the these particular points to this marginal plane.



* still now, we saw that how we can use the support vector classifier & that was specifically for the classification problem.

29th Oct
2022

TOPIC

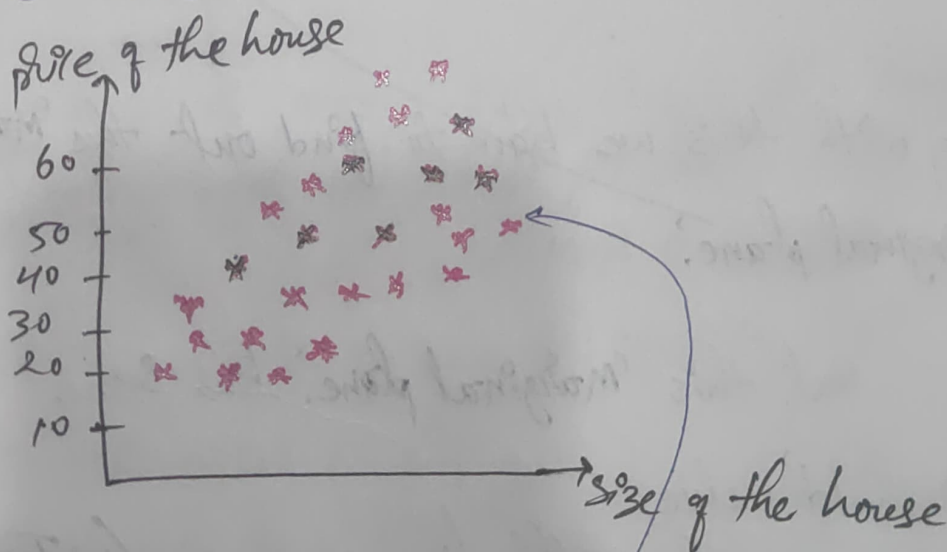
1. SVM : Support Vector Machines
SVR : Support Vector Regression

SVM

(2)

(2) Support vector Regression :-

~~So, if I have~~ Let's say, I have a problem statement, is to find out based on the 'size of the house', I ~~need~~ need to predict the 'price of the house.'

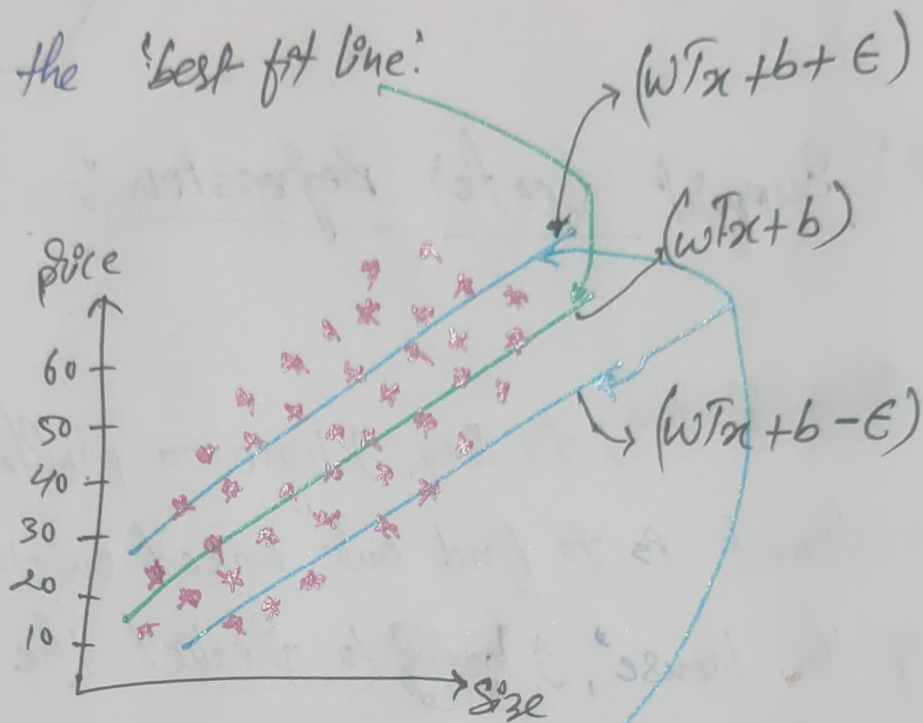


this are the my data points :

Now, what will happen for this particular data points!

How we basically going to use the 'best fit' line & 'marginal plane.'

Our main aim ~~to~~ over here should be that, find out the 'best fit line'.



Along with this, we have to find out the 'marginal plane'.

But, this 'marginal plane' has some 'threshold' value.

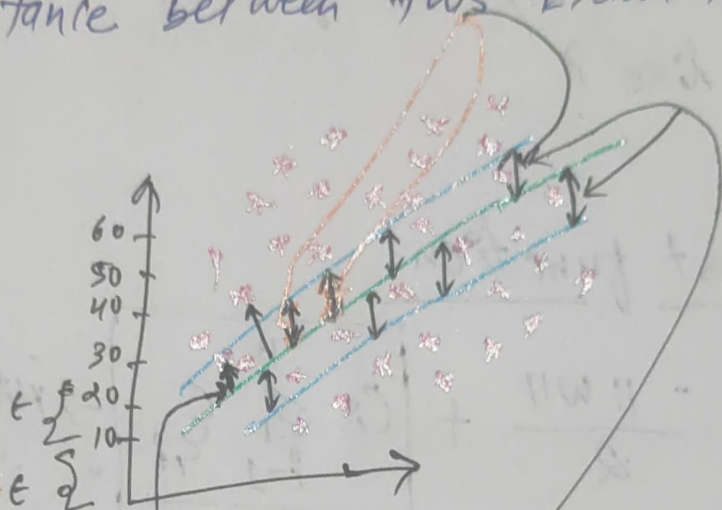
The middle line, gives us a $(wTx + b)$ equation. &

The Top line, I will mention; that in this way $(wTx + b + \epsilon)$

$(\epsilon) \leftarrow$ This 'Epsilon' is nothing but the marginal error.

The below one, it will be the $(wTx + b - \epsilon)$

(ϵ) or 'Epsilon', it means that some 'marginal error' which is basically saying distance between this error to this error.



That basically means, I can create the 'marginal plane', along with the 'best fit line'.

But, I will keep a ~~constraint~~ constrain of ~~this~~ of this (ϵ) Epsilon that this much error is fine.

we can have the point with this much error if we can construct the 'marginal plane'. This is the 'marginal error'.

over here, can we say this ~~are~~ my (ϵ) 'Epsilon' value. ~~the~~ my (ϵ) or Epsilon ^{value} will be somewhere around 5 to 7.

some 5 to 7 (ε) 'Epsilon value' is there on the top of the margin & the bottom margin.

* Now, How my Cost function will look like?

* Cost function

$$\min_{(w, b)} \frac{\|w\|}{2} + \left(C_i \sum_{i=1}^n \xi_i \right) \rightarrow \text{'hinge loss'}$$

my Cost function, will look like this.

Here, I am going to 'minimize' the

(w, b) by changing (w, b) this is nothing

but $\left(\frac{\|w\|}{2} \right) + C_i \sum_{i=1}^n \xi_i$ is obviously

the 'Hyper parameters' C_i & $i=1$ to n .

ξ_i or 'eta of i'

This is equation, this is called

as 'hinge loss'.

Here, I am going to write 'const' 'constraint'.

Constraint:

$$|y_i - w_i x_i| \leq \epsilon + \epsilon_{\eta i}$$

This constraint will change everything.

Here, I am simply saying

$$|y_i \text{ or } y_{\eta i} - w_i x_i \text{ or } w_i x_{\eta i}|$$

This means the difference between the 'truth points' & the 'predicted points'.

Should be ($\leq$) less than or equal

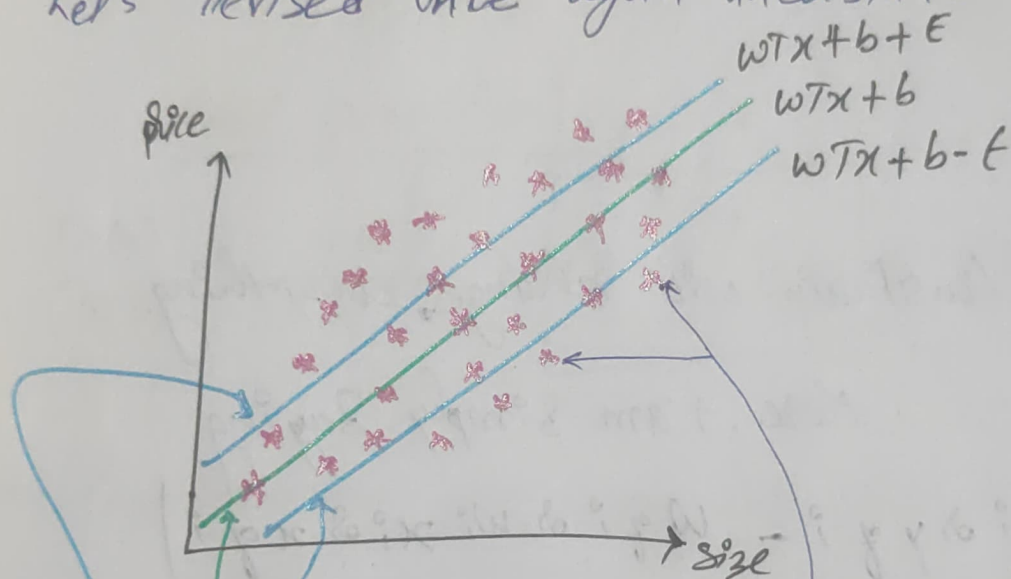
to (ϵ) or Epsilon + $\epsilon_{\eta i}$ or Eta of i .

40:49

* Let's understand :-

$$\epsilon + \epsilon_i$$

Let's revised once again understand it.



Let's say this are my data point.

my main here is to find out the 'best fit line', along with that, I need to find out the marginal plane.

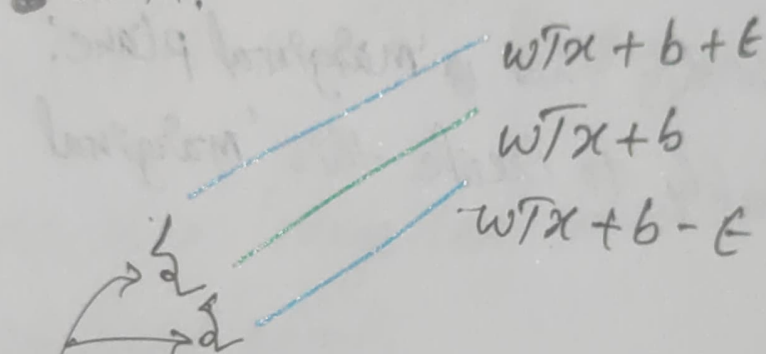
* In Regression, we will not going to find the right value. There will be always some difference, because this is a continuous value.

So, we can't talk 'incorrect points' over here.
Instead of that, I can talk about
the (ϵ) or 'Epsilon'.

Let's say,

I am equation, this particular

with.



why . up line is (+)ve (ϵ) or Epsilon?

Because, (ϵ) ; I am considering

this line as my ~~some~~ distance.

within this distance is coming
my 'marginal plane' is perfect.

(ϵ) or Epsilon is nothing but my
~~that~~ 'marginal Error'.

This 'marginal Error' nothing but
like a 'hyperparameter'.

$\epsilon \rightarrow$ marginal Error
 $\downarrow$
 Hyperparameter

'Hyperparameter' basically means even though if I am getting this Error.

within this particular points are falling within this 'marginal plane'. I am OK, probably to create this 'marginal plane'.



* Cost function :-

$$\min_{w, b} \frac{\|w\|^2}{2} + \left(\sum_{i=1}^n |\epsilon_{\tau_i}| \right)$$

$\nwarrow$
 Hinge Loss

So, the 'cost function' here would be the
(min) minimizing the (w, b) mode of

$\frac{\|w\|}{2}$ + . In Hinge loss we have
the C_i or $C q_i$ & then $\sum_{i=1}^n$ or Submission
of $(i=1)$ to n . & ϵ_i or Eta of i .

This is a 'Hinge loss' which we have
discussed about.

* Since initially we have added the
 ϵ_i or Eta, so, what will be the
constraint? If we have ~~use~~ use the
Hinge loss?

* Constraints :-

$$\bullet |y_i - w_i x_i|$$

If I will write the w_i of x_i , Here
basically I am trying to find out the

$$|y_i - w_i x_i|$$

difference between the 'Truth' & predicted
point.

~~* what will be~~

* with respect to the 'size of the house'
or
new size

~~what will be my price of the house~~

what will be my 'price of the house'?

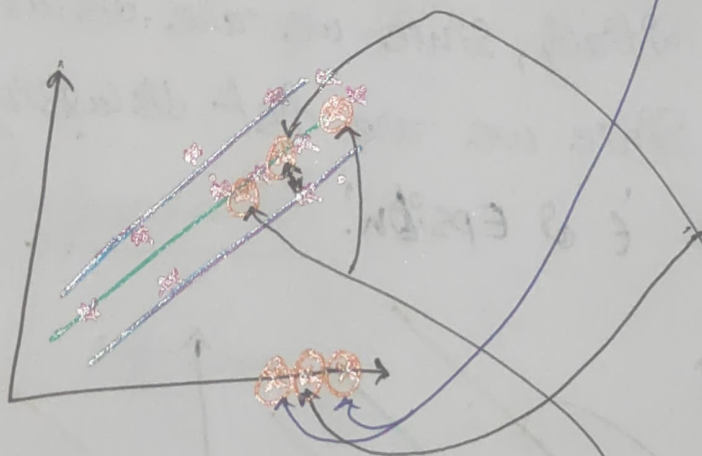
that is what we are predicting?

Let's say,

If I get a new data point,
what will be the output over here.

1st Constraints:

Let's say, If I get a new data points.



The output will be coming over here.

So, Here I really need to find out the
distance between the 'truth' & 'predicted.'
I will be able to calculate.

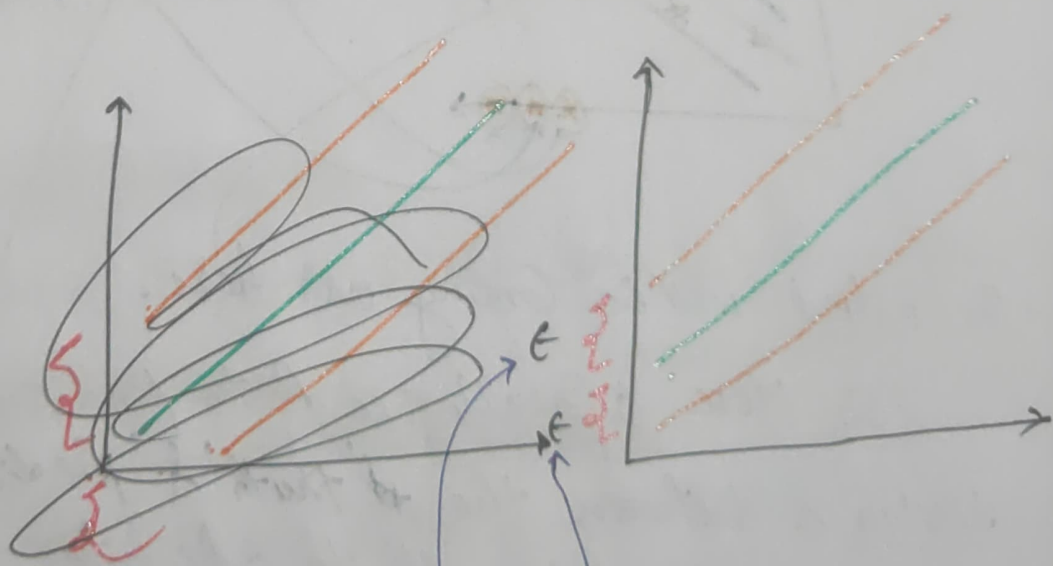
Let's say,

with respect to this data point,
I am trying to calculate the output.

This is $y_i (y_i) - w_i x_i$
 $x_i (w_i x_i)$.

$$* |y_i - w_i x_i| \leq \epsilon$$

Now the constraints should be that the difference between these 2 should be ($\leq$) less than or equal to ' ϵ or Epsilon', ~~the 20%6~~ initially, since we are discussing. Since we are just discussing about the ' ϵ or Epsilon'.



* what is ϵ or Epsilon?

It should be less than this particular marginal plane only!
This is also (ϵ) & this is also (ϵ)

So, we are agree with the 1st constraints.

If we are able to bring this distance less than this (ϵ) or Epsilon.

That means, my model is performing well. Because, it is being able to give an error, which is less than this 'Error'.

This is my 1st constraint.

* 2nd Constraints

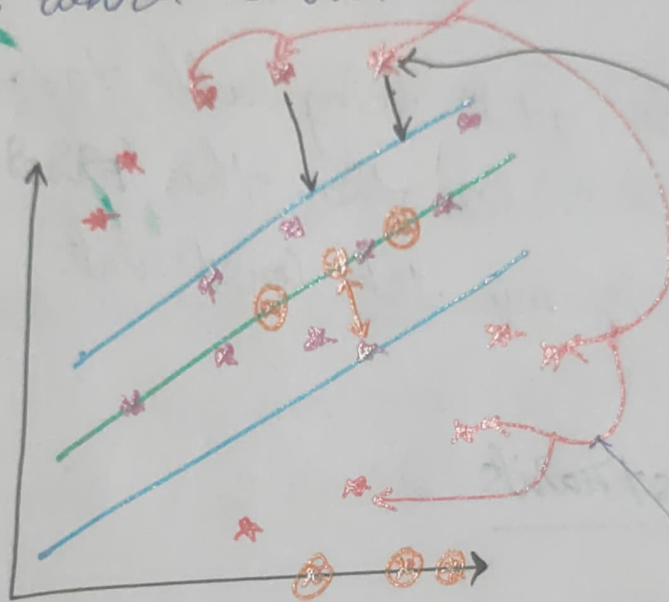
Let's say

* 2nd Constraints :

$$|y_i - w_i x_i| \leq \epsilon + |\xi_i|$$

Let's say,

I have some other data point, which is over here.

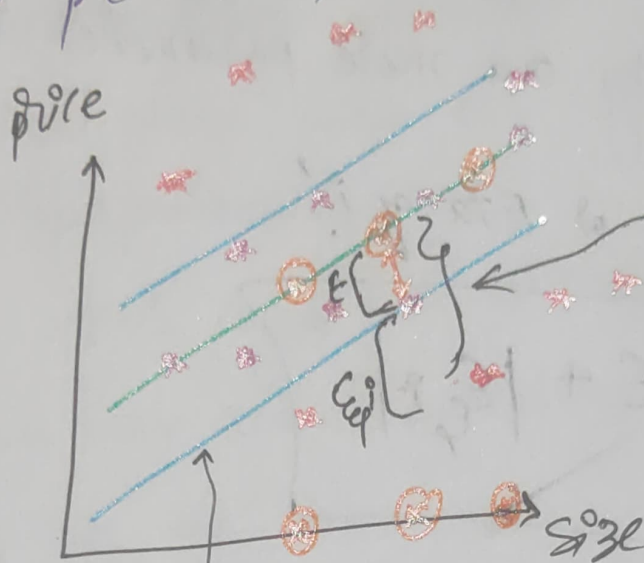


If I have this particular data points, this also needs to be treated.

This is treated with this particular value that is ' ξ_i ' or $\epsilon + \eta$!

Now, ξ_i basically means, what is distance of the this particular point to this particular 'marginal ~~plane~~ plane'.

* ~~and~~ what we will be doing is that, any prediction, we do.



Let's say, this prediction, we do, we are going to take 2 condition.

one is the 'epsilon' distance & along with it, we are going to add one more distance that is ϵ_i & η_i .

I am saying that, ok even though within this particular distance, if we are falling with respect to the predicted & real point.

I am ok to consider this as on my 'marginal plane'

* what, I was doing was,

~~doing~~ Along with ' ϵ or Epsilon',
I am adding one more parameter that
is the ' ξ_{ϵ} or Eta of ϵ '
made of

$$\epsilon + |\xi_{\epsilon}|$$

~~to~~ I am just putting this particular
condition.

what is this

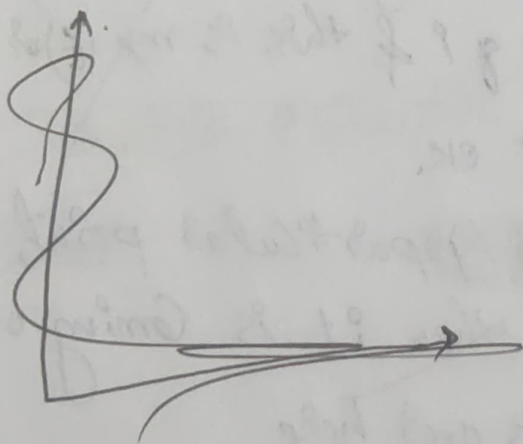
* ϵ or Epsilon : This is nothing but
my 'marginal Error'.

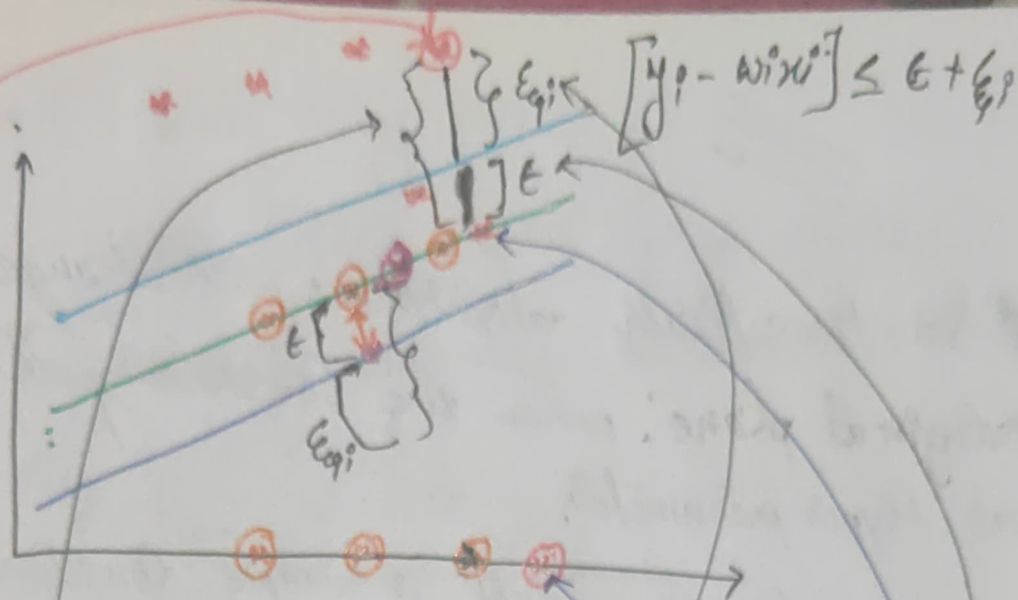
* ξ_{ϵ} or Eta of ϵ : It is nothing but 'Error'
above the margin.

* Based on these 2 parameters, we can
create best marginal line. marginal line.

it automatically algorithms will change the 'marginal plane', when we considering this as Hyper parameter.

this is just a simple constraints that we have added, remaining all the things are the same.





This value, we will be seeing over here,
 this is my (ϵ_i) or ϵ of i & this is my (ϵ) or Epsilon
 I am saying that, OK.

For this ~~particular~~ particular point, if
 I have to predict it, then it is coming over here
 but my real output is over here.

So, the distance between these 2
 $(\epsilon_i \& \epsilon)$ should be less than that is
 $[y_i - w_i x_i]$ should be less than ($\leq$)
 $\epsilon + \epsilon_i$.

If it is satisfying this, then I don't
 have to change my 'marginal plane'.

* They can also be point above the (ϵ_i) .

If the distance is very very high which is not within the ~~hyper plane~~ 'hyper parameter' then we have to change this marginal plane.

* In short this all things, will help us to basically setup the particular marginal plane.

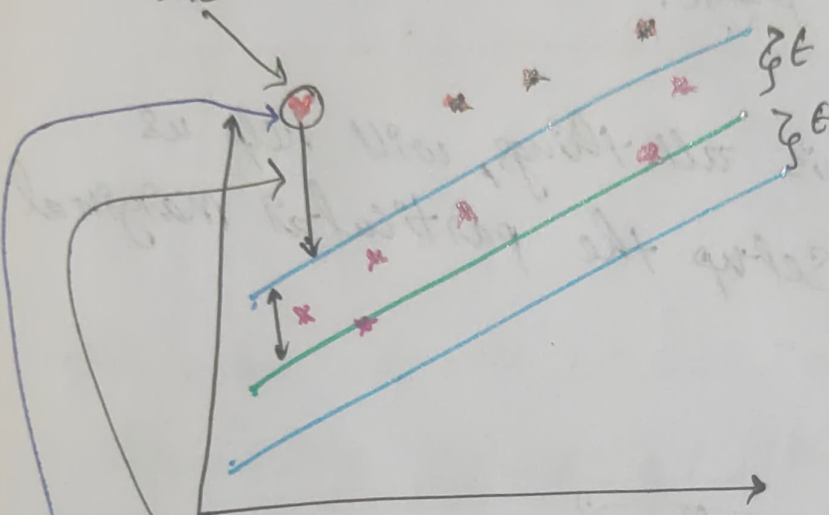
* ~~Scenario & Question?~~

~~What.~~

* Question :-

what if my ϵ_i or $\text{Eta } \eta_i$ value is very very Big? then what will happen to this data point?

$$\epsilon_i > \text{Big}$$



If this data point is also there, then my marginal plane line will not change.

Because this distance is is very

high.

Along with this ϵ_i or $\text{Eta } \eta_i$ is also there.

So, we are always going to get this value

$$\epsilon + \epsilon_i$$

less than or equal to this.

But this is wrong.

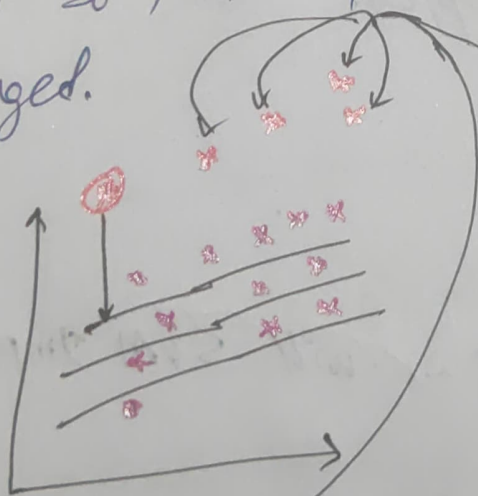
It should not happen.

So, we performed this as a ~~hyperparameter~~
hyperparameter.

$$\epsilon + \epsilon_y i$$

So, we perform this as a hyperparameter task.

we have to continuously adjust this value with the (ϵ) or Epsilon to make sure that, our points are very near to each other. So that my 'marginal plane' does not change.



In this particular case, since I have an outliers & some other points over here then my 'marginal plane' will keep on changing.

* Question :-

How long, I will continue adjusting the marginal plane?

we don't have to worry about the continue the set up the 'marginal plane' that we can basically perform, with the help of 'Hyper parameters' ~~and~~ Tuning.

wherever the performance metric is better for with respect to the training dataset, then we stopped there.

* Question :-

~~SVM~~ SVM outliers - will SVM impacted by the outliers?

Yes, obviously it will going to impacted by the outliers.

Question :

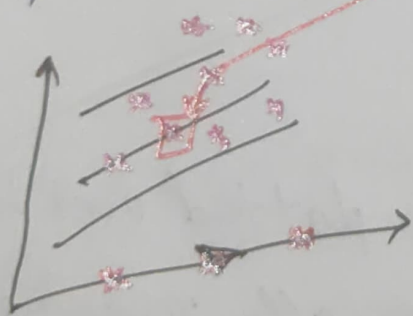
do we need to Normalized our data points? like standardization?

yes, we need to perform it.

Question :

How to predict in 'SVR'? not in SVM?

whenever with respect to new data points, it will try to predict over here.



This will be my predicted point.

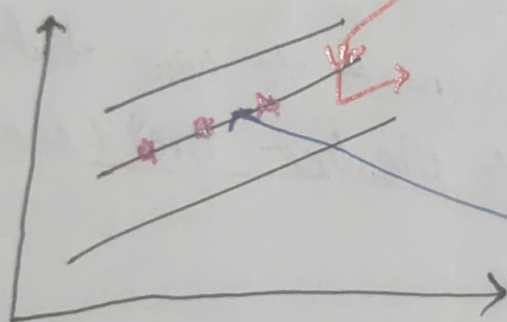
* It involve distance, so we need to ~~standardized~~ standardization or normalized it.

* Question :-

why $(y_i - w_i x_i)$?

' $w_i x_i$ ' is nothing but $\hat{y}$ or $\hat{y}$ hat

this $(y_i - w_i x_i)$ is nothing but
this is my equation for this line



whatever point will be plotted over here
that is what it is basically meaning.
that is my real output.

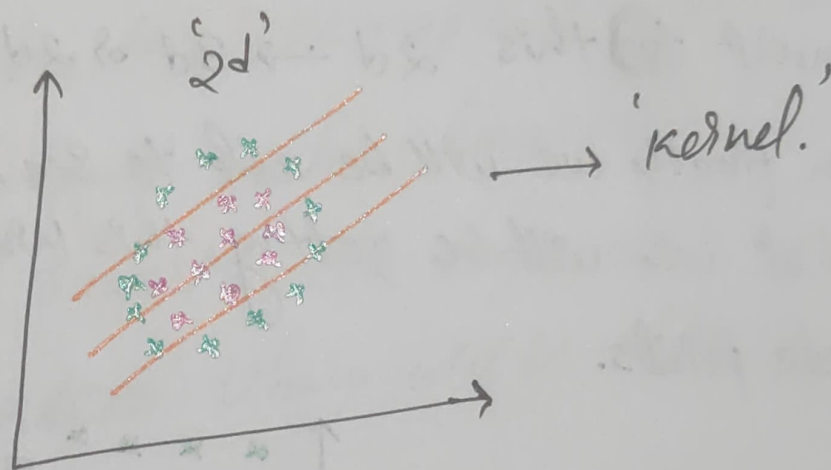
$$|y_i - w_i x_i|$$

this is basically a 'Loss function' or
MAE (Mean Absolute Error).

SVM kernel

Let's say,

I have a dataset, why this SVM kernel is super important.



Let's say, I have a dataset, which looks like this.

Do we can solve this problem, with the SVM or SVR?

Let's consider with 'SVC', we will get the 50% accuracy.

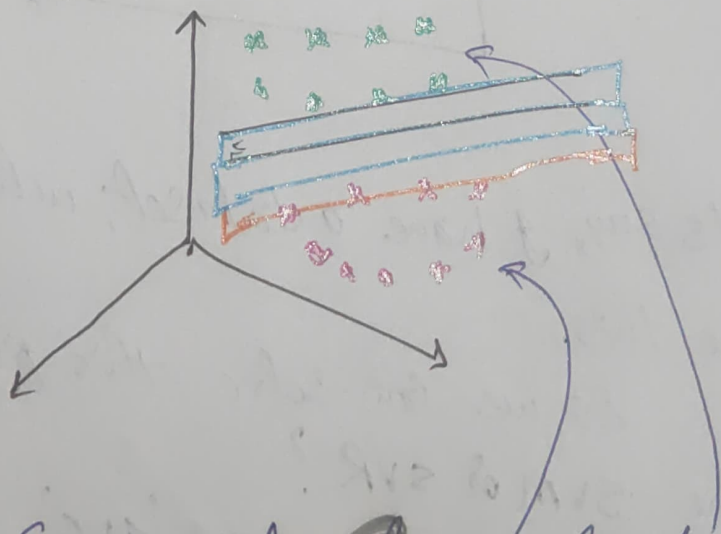
So, ~~in that~~ this case, we use the 'kernel'.

* 'kernel' is nothing simple transformation technique.

Can we say this all points in a '2d.'

If we do a transformation, where we convert this 2d $\rightarrow$ 3d & 2d into 3d.

then finally we will be able to see, that we will be getting this kind of data points.



we can see green points, somewhere here & another is somewhere here like this.

* we have converted 2d into 3d, we raised the Δ Bar. & then probably we can apply a Δ marginal plane, it will come.

Here a another 'marginal plane' also come,
over here. & we will be able to divide this.

_____ x _____ x _____ x _____

practical Now 5

* SVC practical example & SVR.

SVC: (Support vector classifier)

SVR: (Support vector Regressor)

'sklearn' library will be there.

* Tomorrow Agenda

* SVM kernel, ROC & AUC curve

_____ x _____ x _____ x _____

2:45 to 7:00. practical start